



US 20250283463A1

(19) **United States**

(12) **Patent Application Publication**
Mullins et al.

(10) **Pub. No.: US 2025/0283463 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL**

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F04B 47/00 (2006.01)

F04B 53/02 (2006.01)

F04B 53/22 (2006.01)

F16J 15/02 (2006.01)

F16J 15/16 (2006.01)

F16J 15/18 (2006.01)

(52) **U.S. Cl.**

CPC **F04B 53/164** (2013.01); **F04B 37/12** (2013.01); **F04B 53/22** (2013.01); **F16J 15/028** (2013.01); **F16J 15/166** (2013.01);

F16J 15/18 (2013.01); **F04B 15/04** (2013.01);

F04B 47/00 (2013.01); **F04B 53/02** (2013.01)

(21) Appl. No.: **19/218,880**

(22) Filed: **May 27, 2025**

Related U.S. Application Data

(60) Division of application No. 17/890,818, filed on Aug. 18, 2022, now Pat. No. 12,345,253, which is a division of application No. 17/308,163, filed on May 5, 2021, now Pat. No. 11,421,679, which is a continuation-in-part of application No. 16/916,657, filed on Jun. 30, 2020, now abandoned.

Publication Classification

(51) **Int. Cl.**

F04B 53/16 (2006.01)

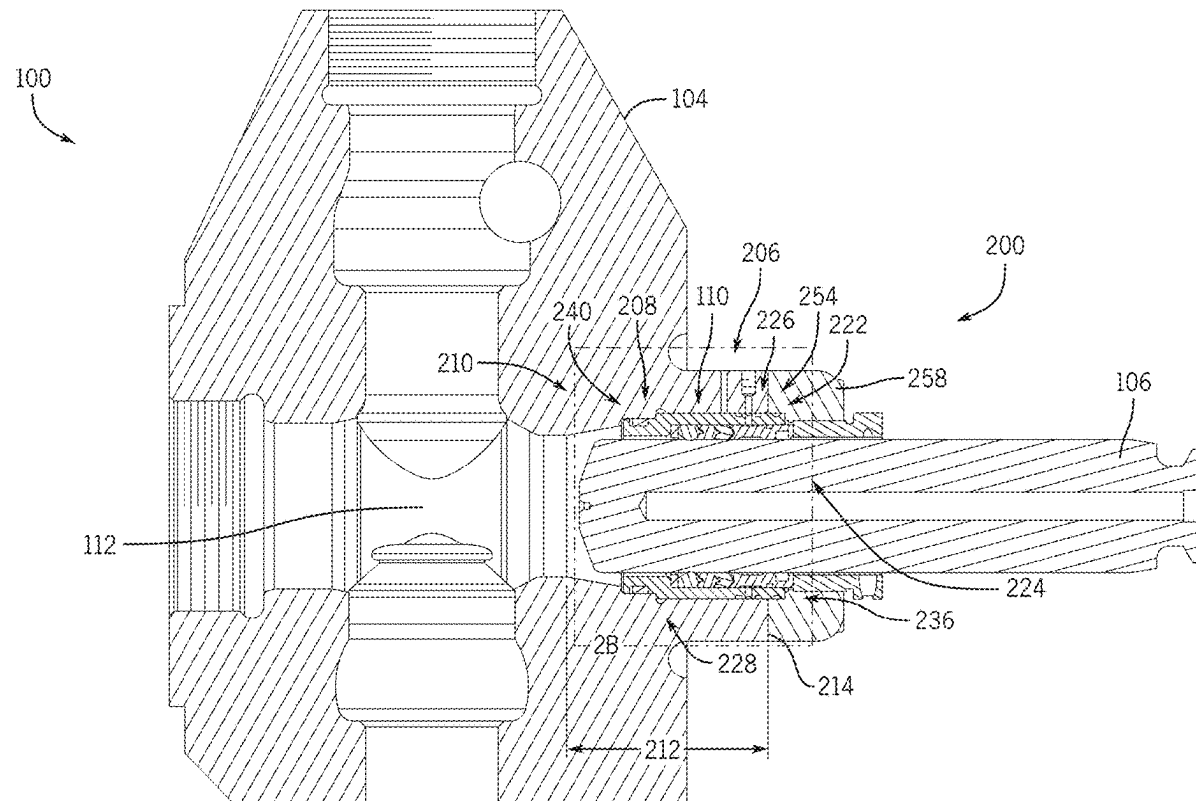
F04B 15/04 (2006.01)

F04B 37/12 (2006.01)

(57)

ABSTRACT

A packing assembly includes a packing sleeve having a first sleeve portion and a second sleeve portion, wherein a first sleeve portion outer diameter is larger than a second sleeve portion outer diameter and a first sleeve portion inner diameter is larger than a second sleeve portion inner diameter. The packing assembly also includes a groove formed in the second sleeve portion outer diameter. The packing assembly further includes a shoulder formed at a transition between the first sleeve portion and the second sleeve portion. The packing assembly includes a shelf formed at the transition between the first sleeve portion and the second sleeve portion. The packing assembly also includes a plurality of packing components arranged within the first sleeve portion inner diameter, at least a portion of the plurality of packing components engaging the shelf. The packing assembly further includes a seal positioned within the groove.



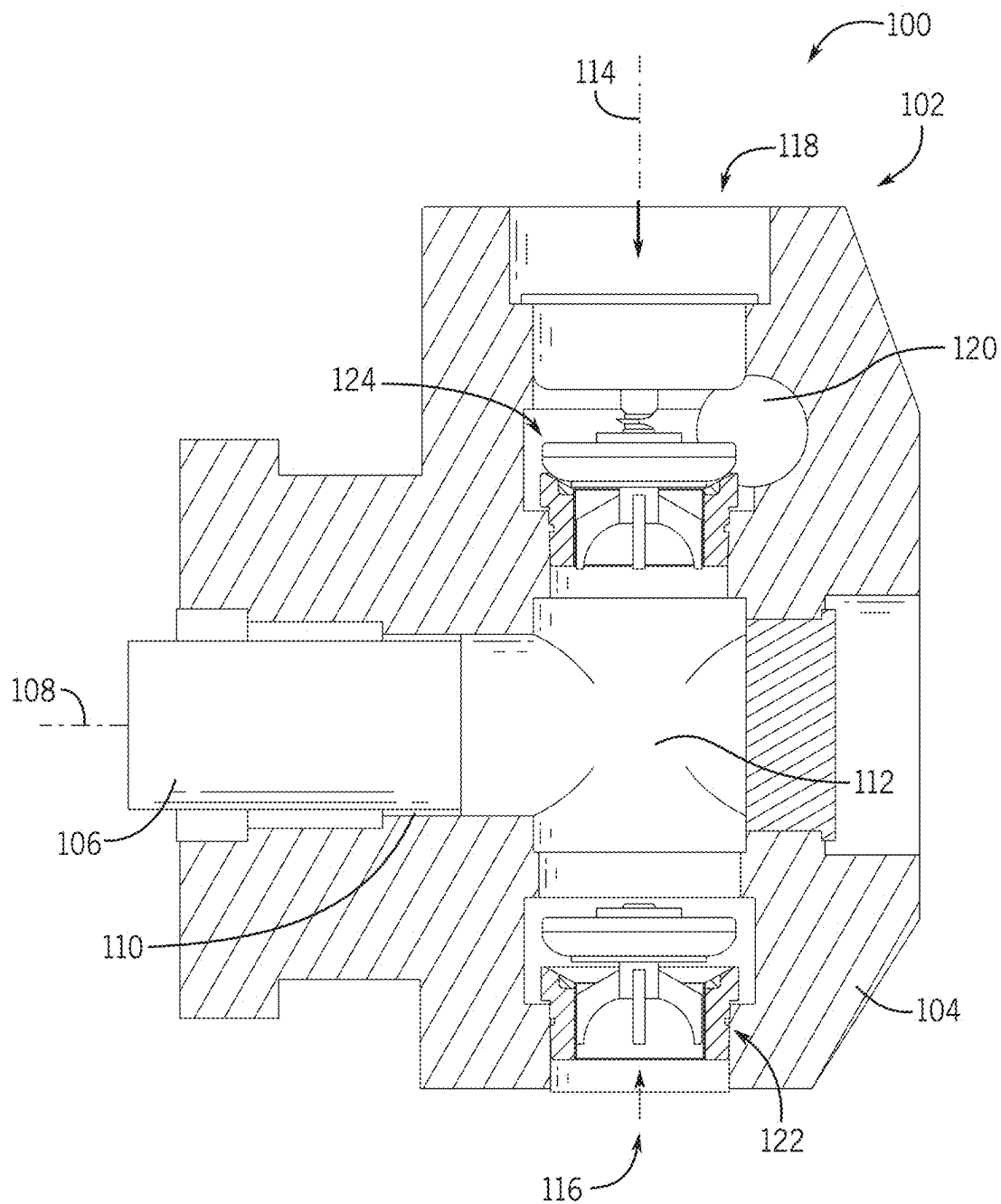


FIG. 1

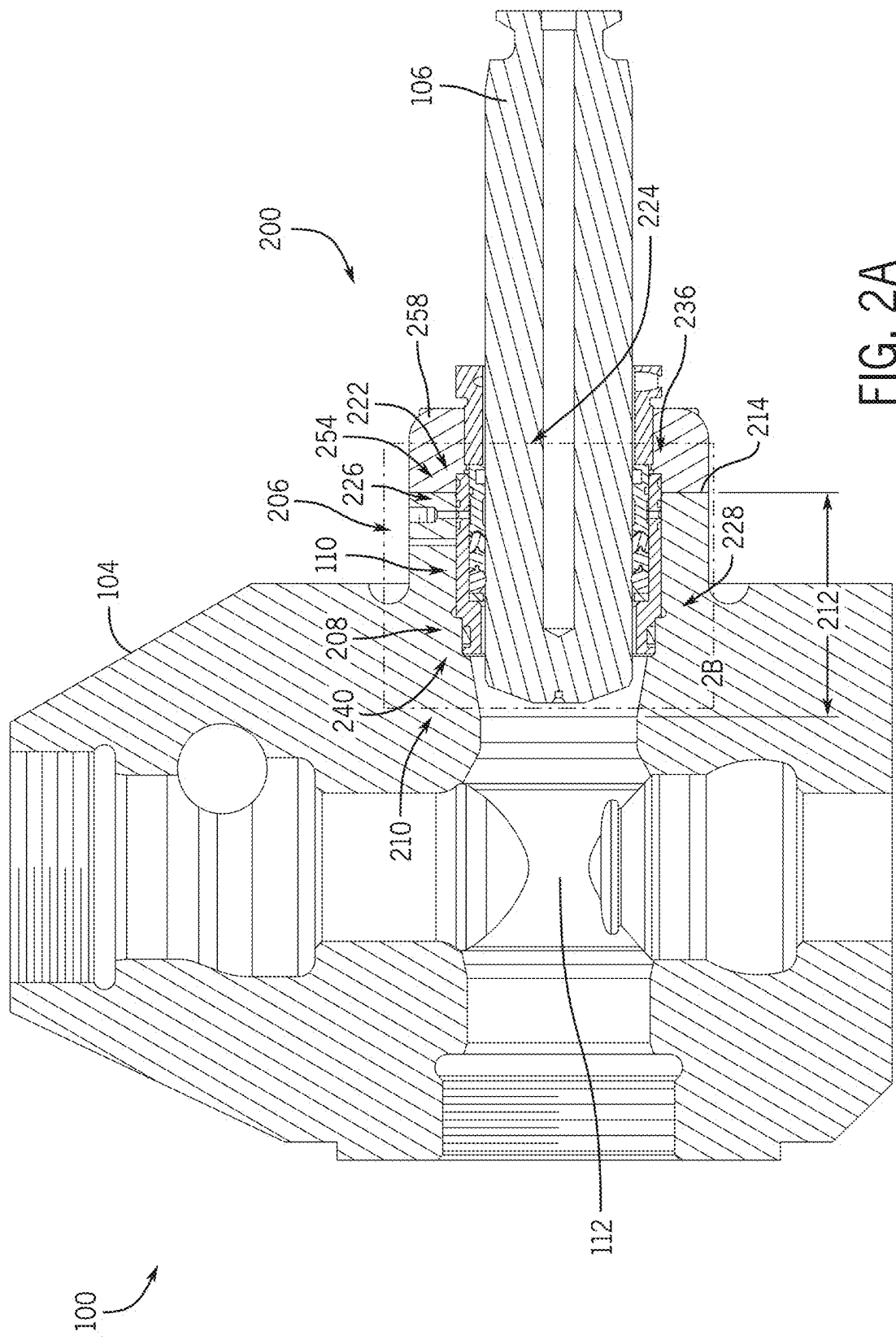


FIG. 2A

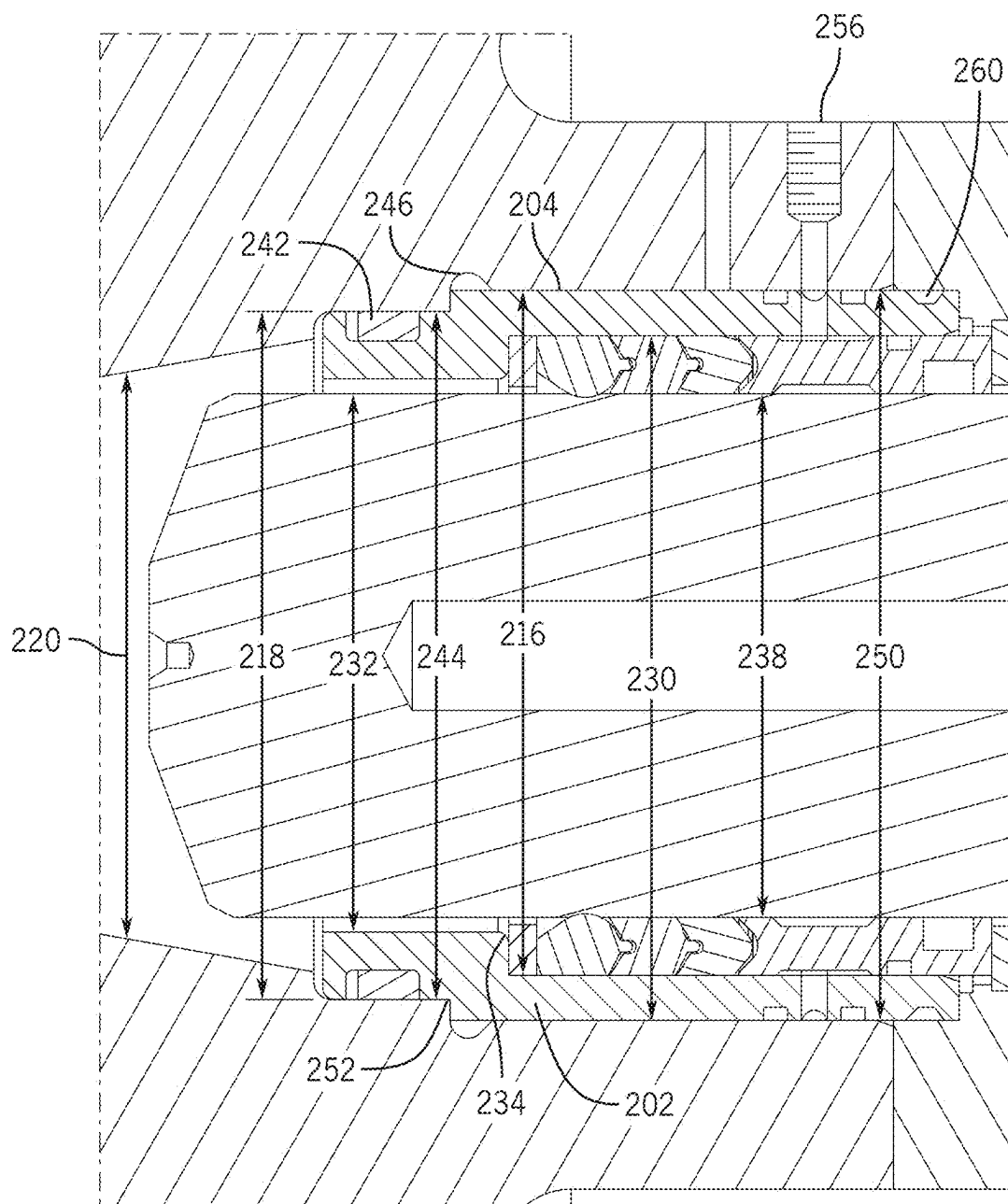


FIG. 2B

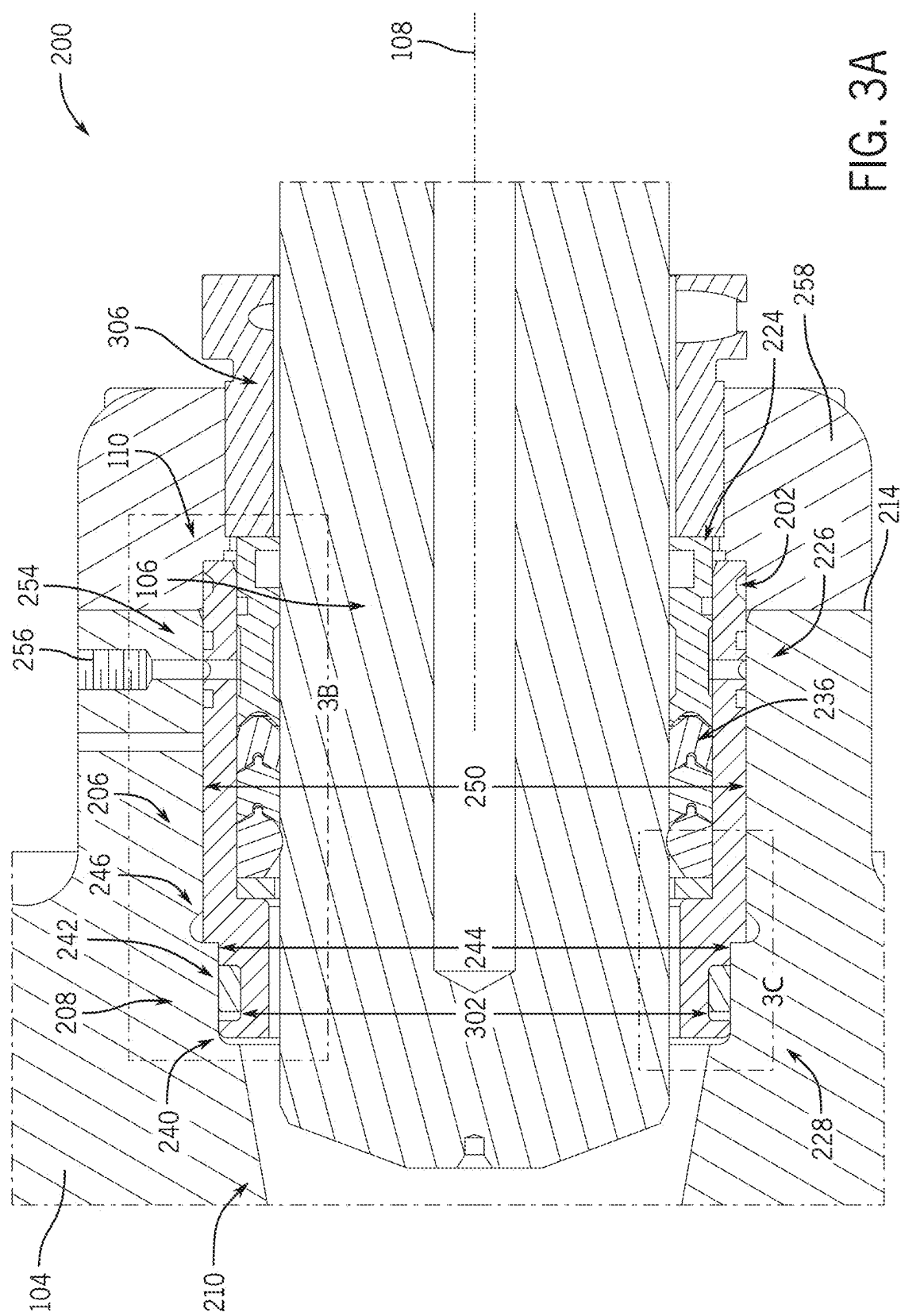


FIG. 3A

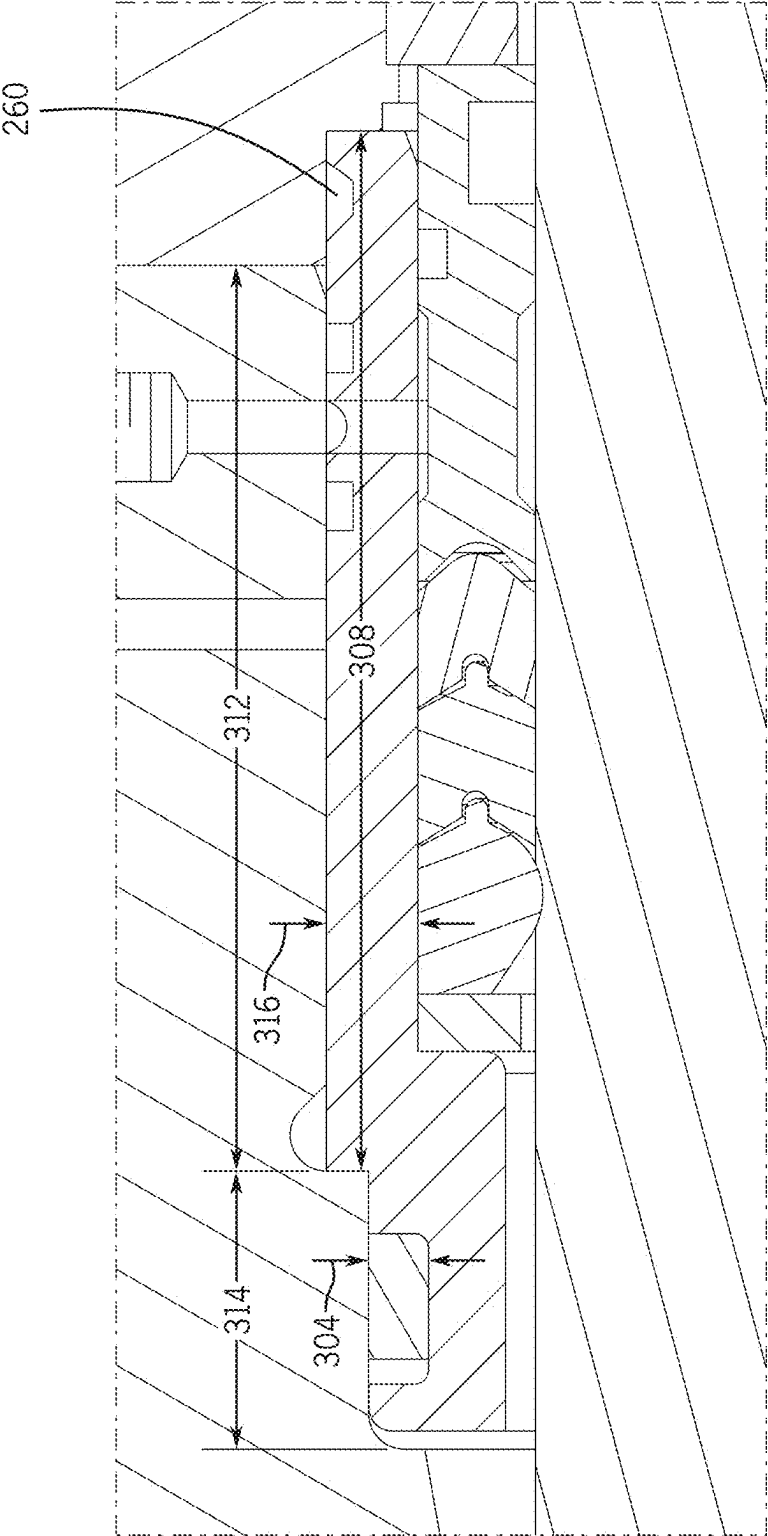


FIG. 3B

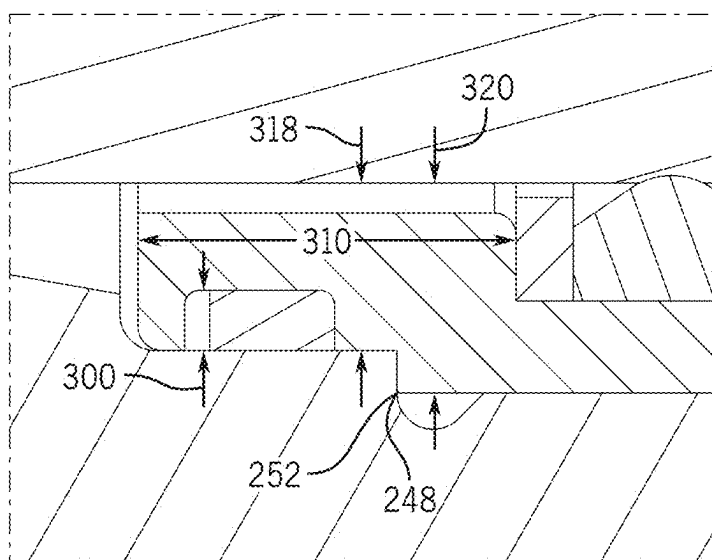
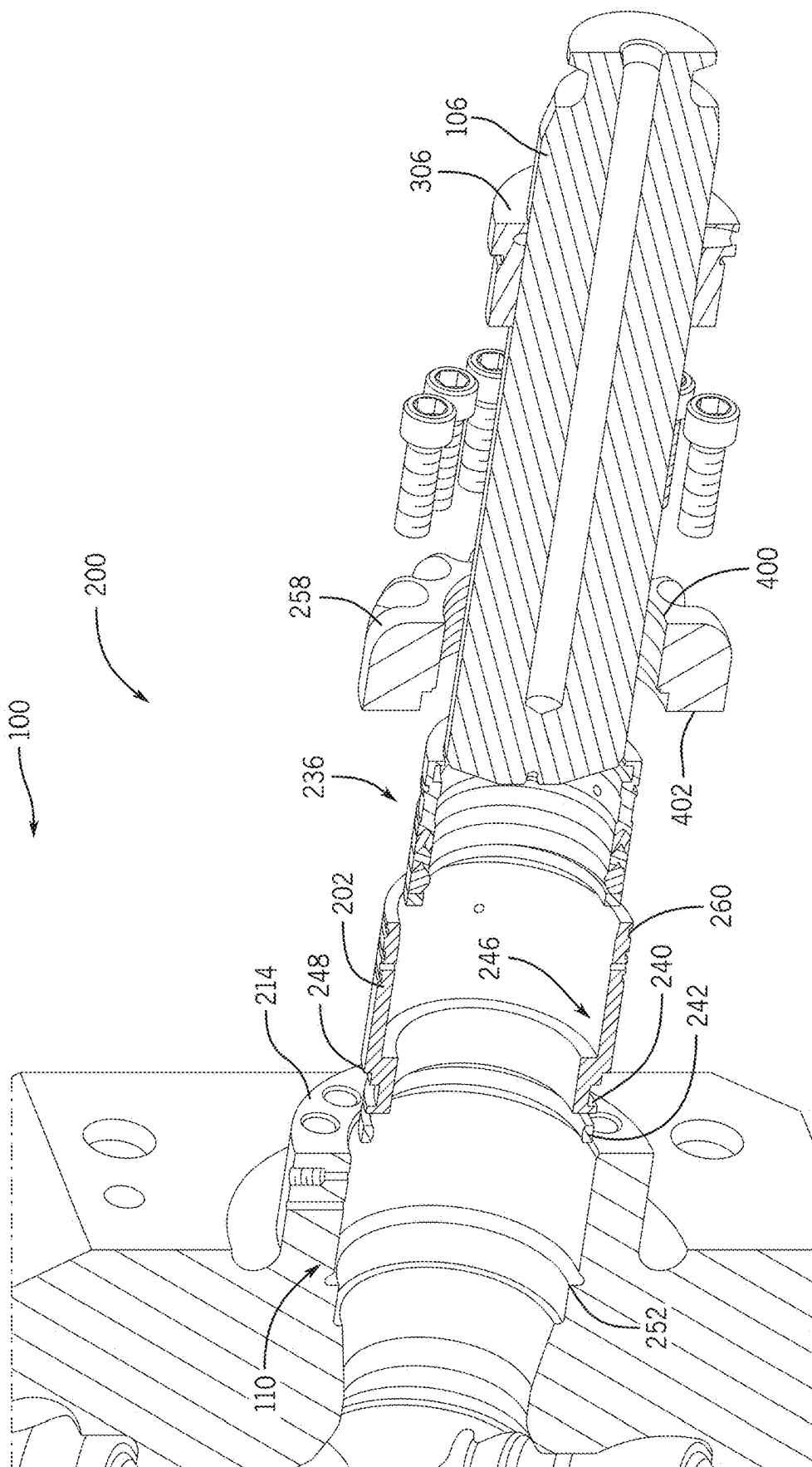
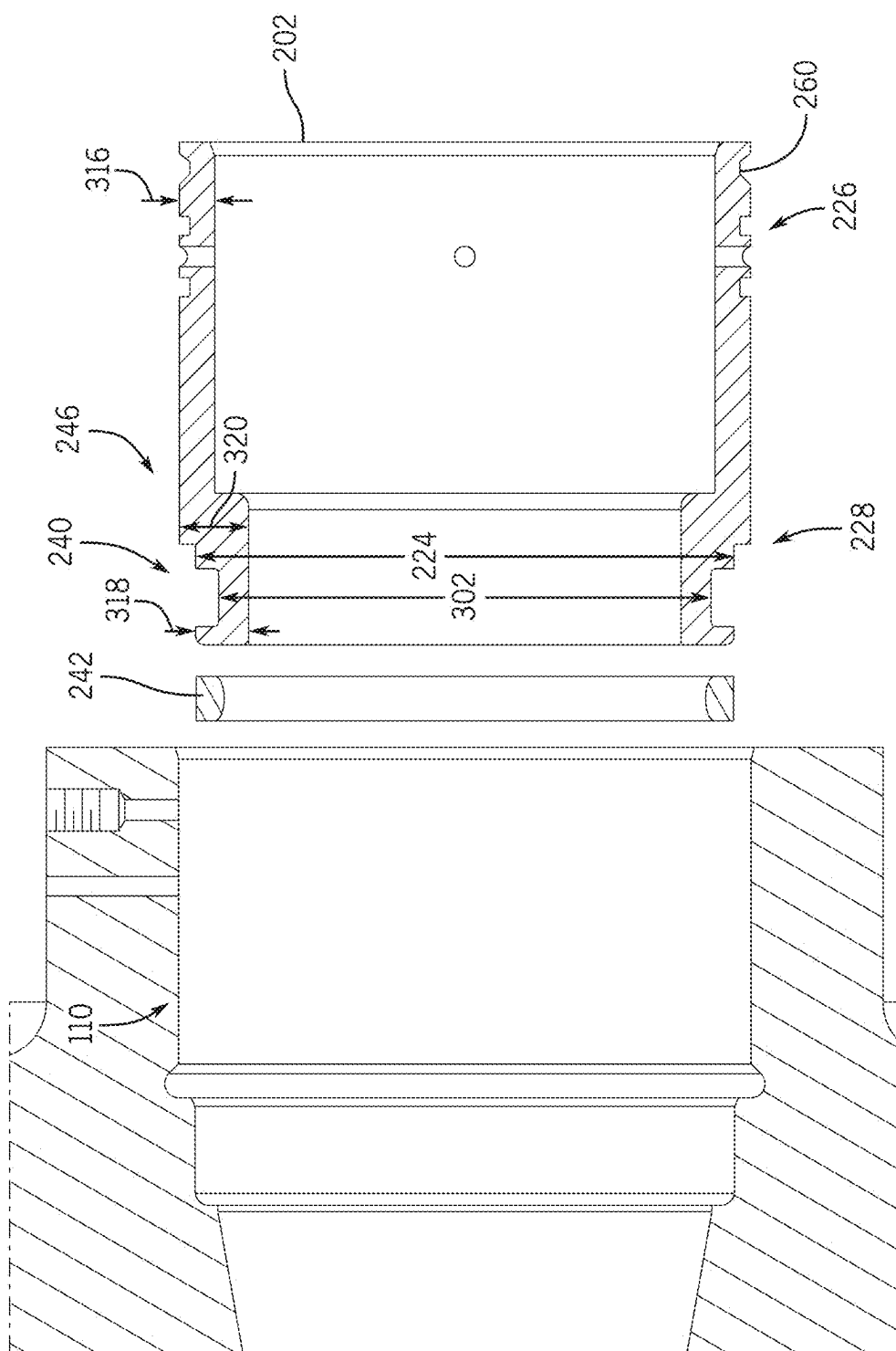


FIG. 3C





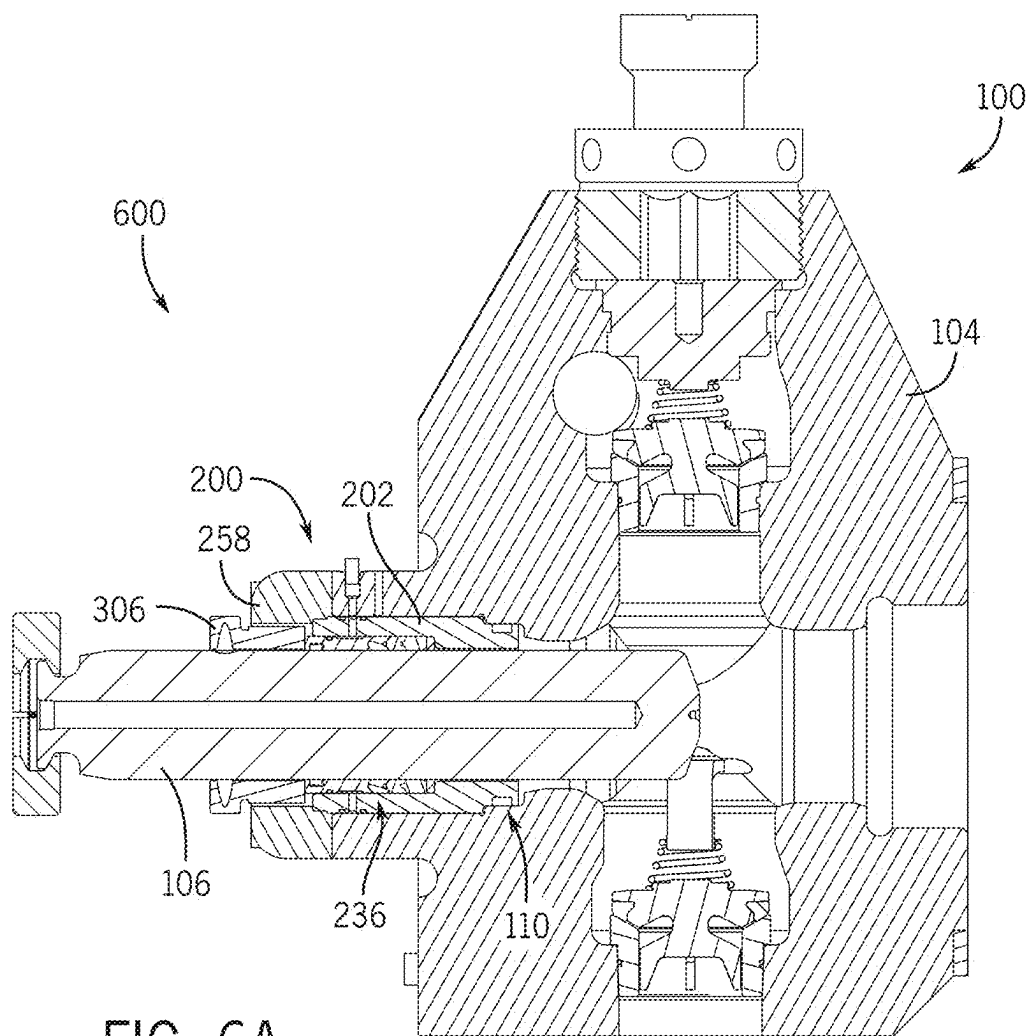


FIG. 6A

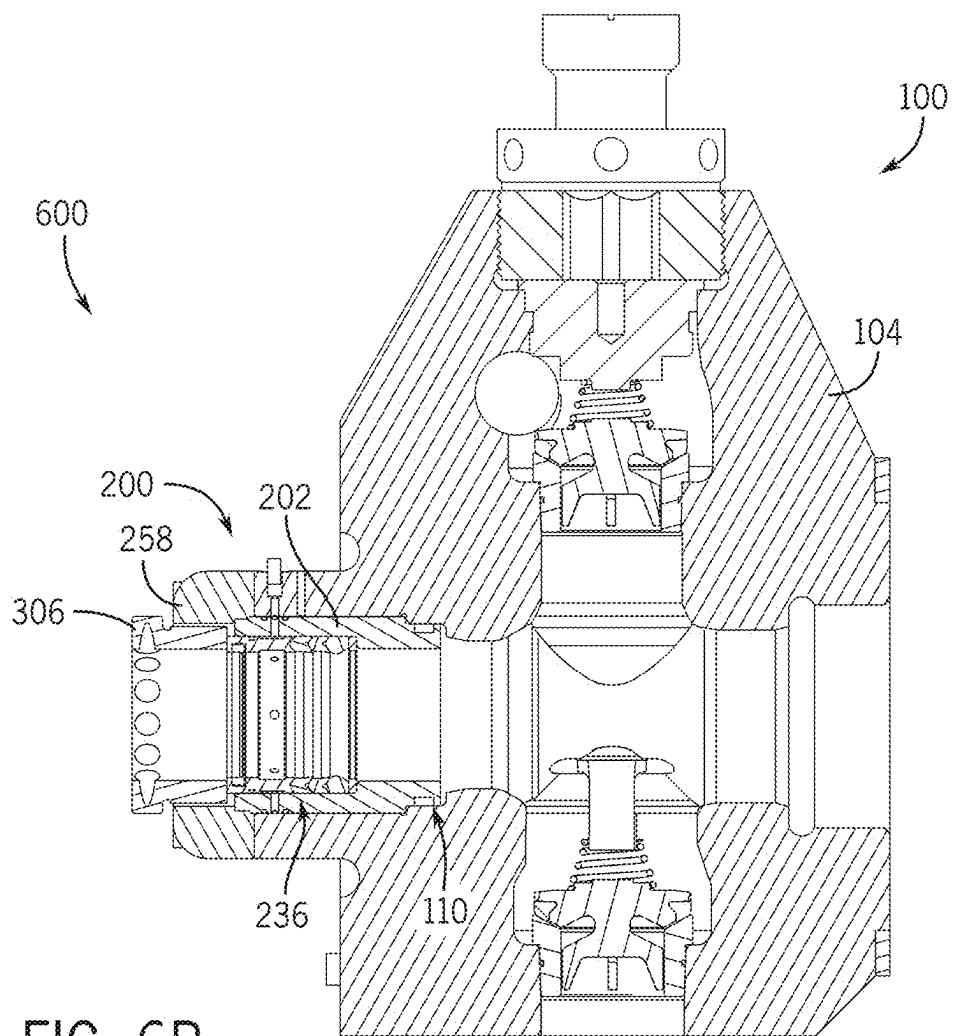


FIG. 6B

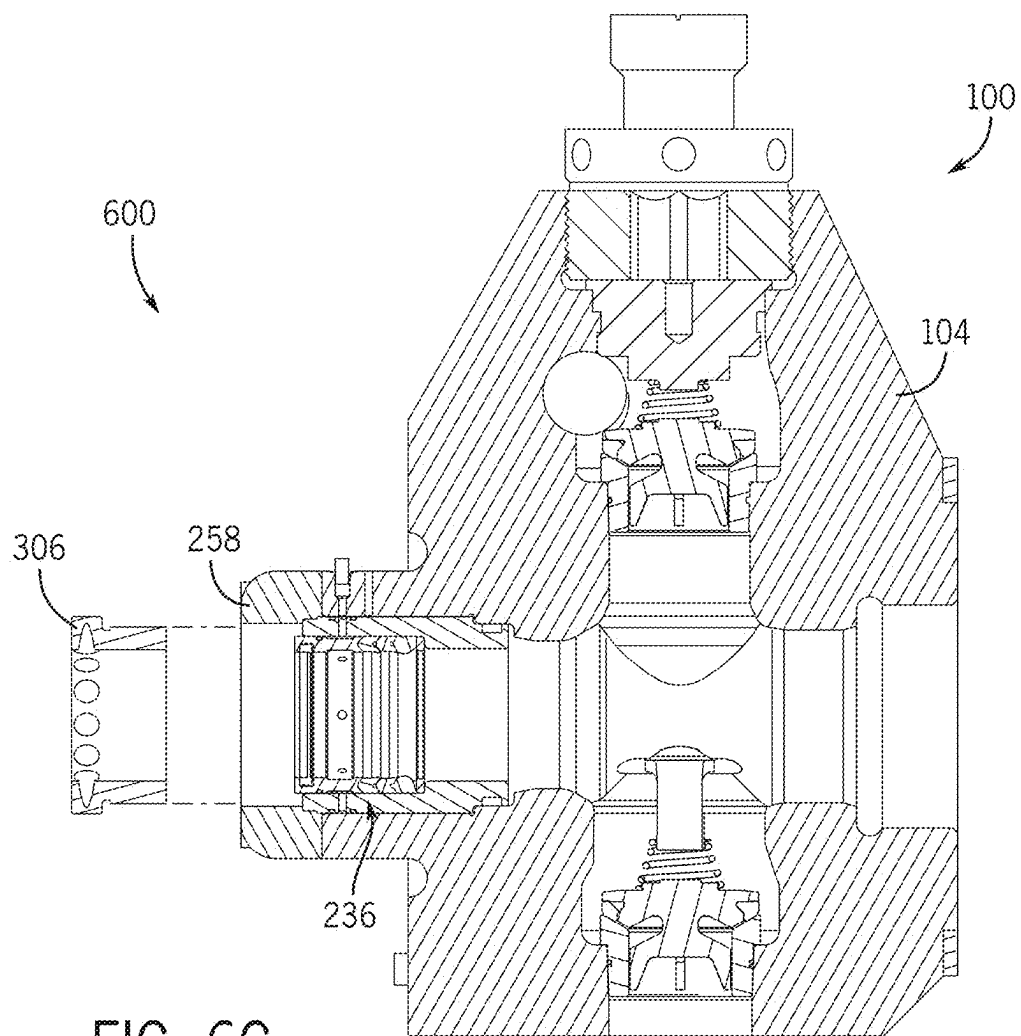


FIG. 6C

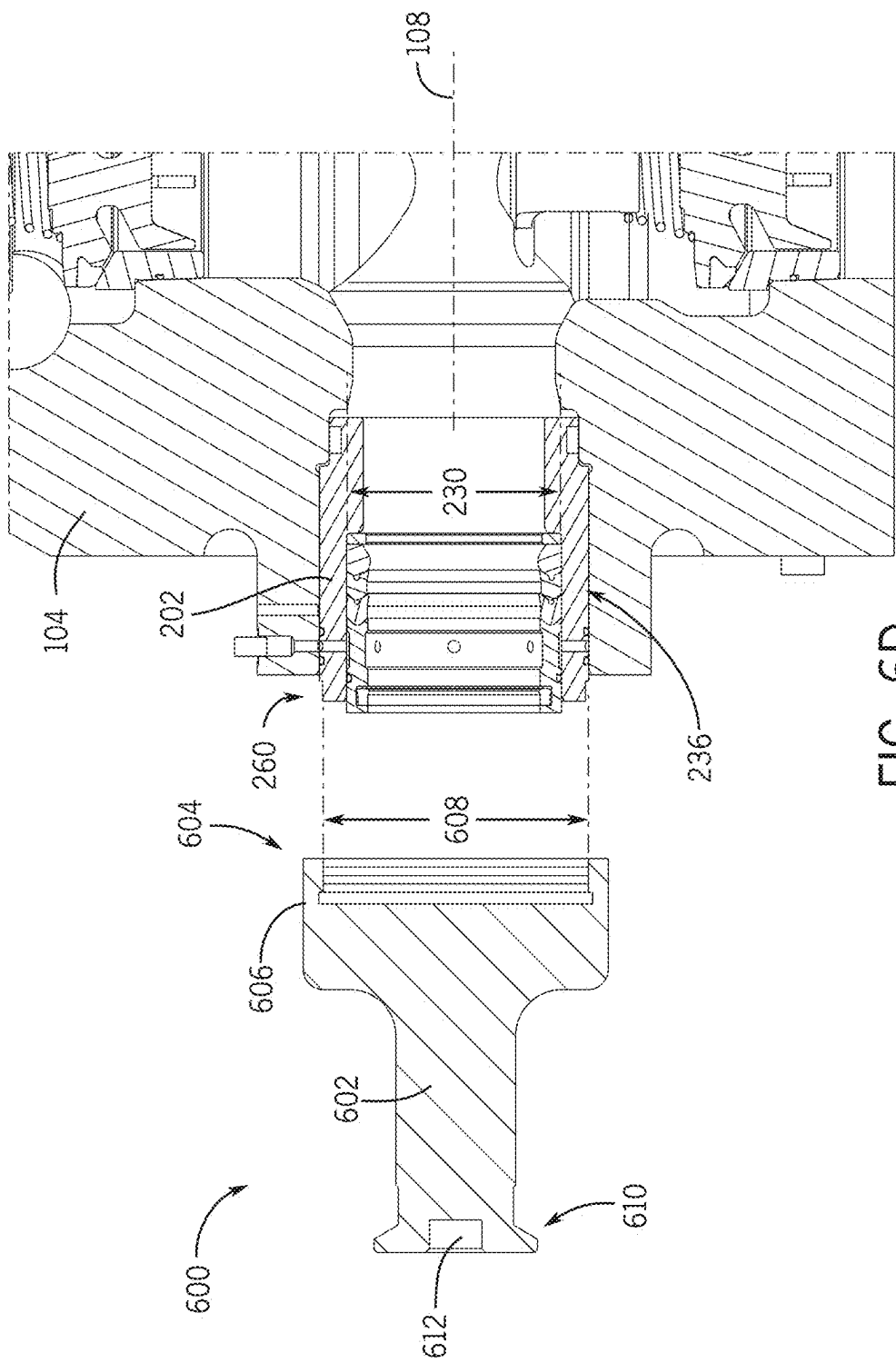


FIG. 6D

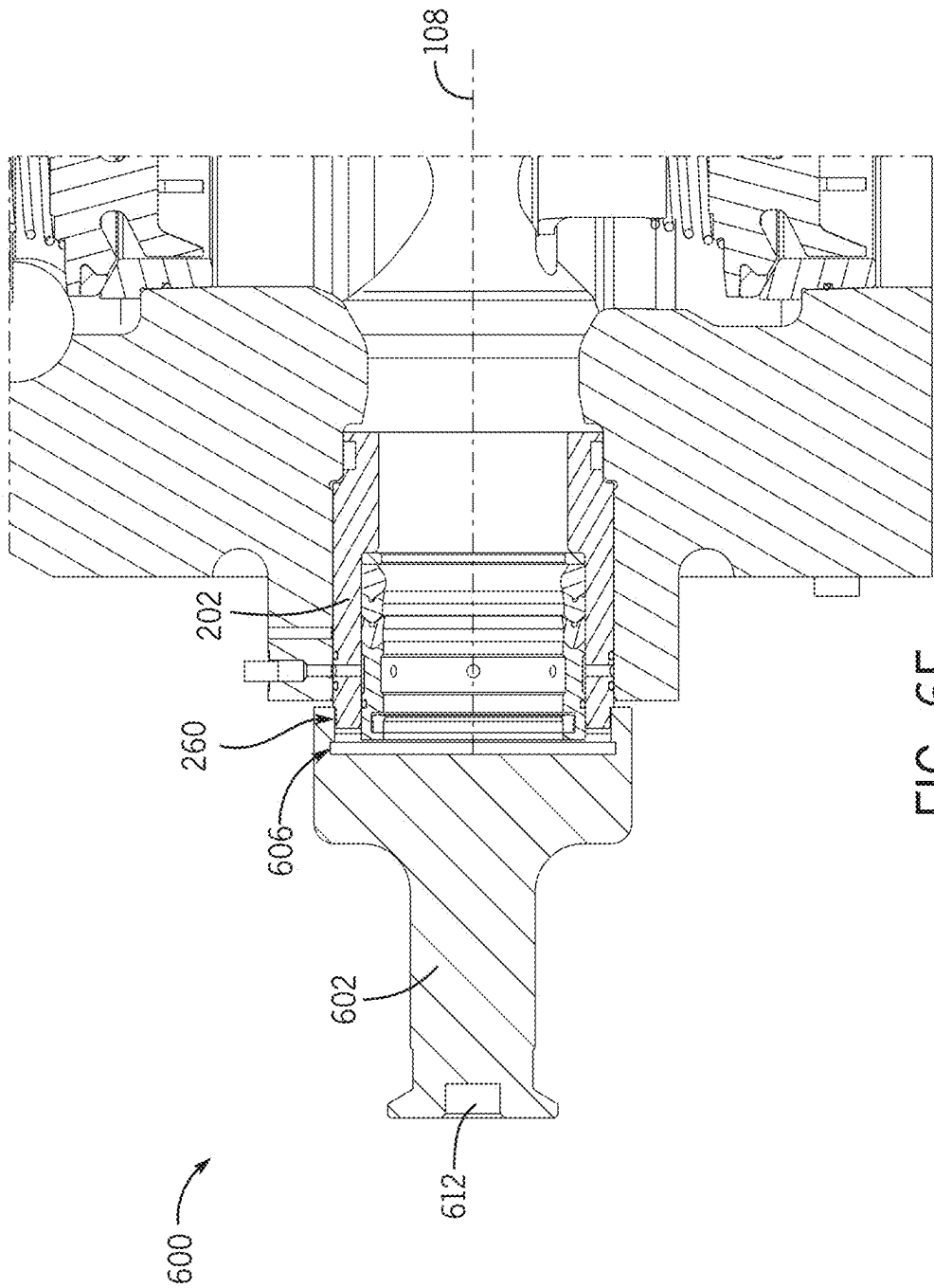


FIG. 6E

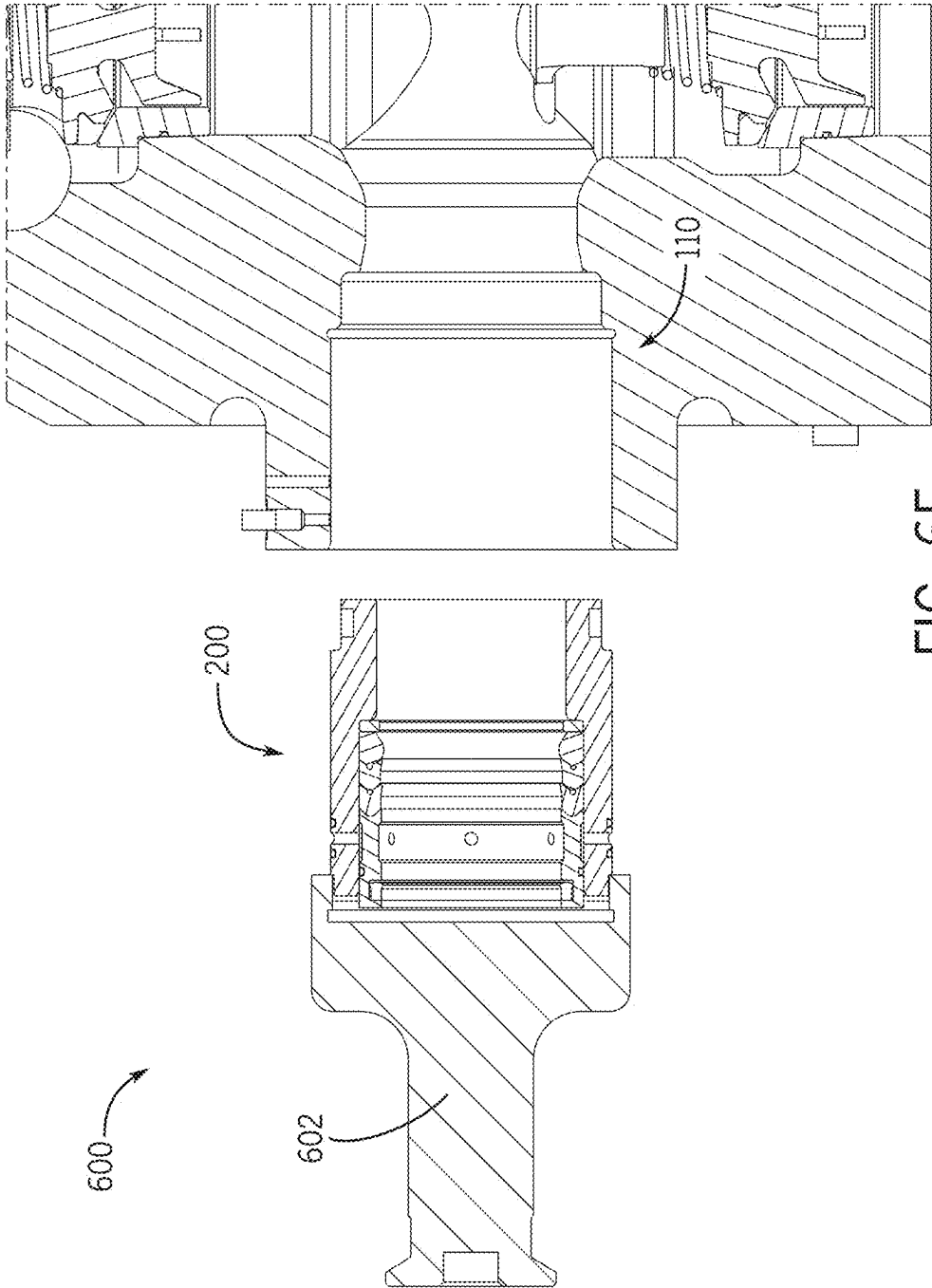
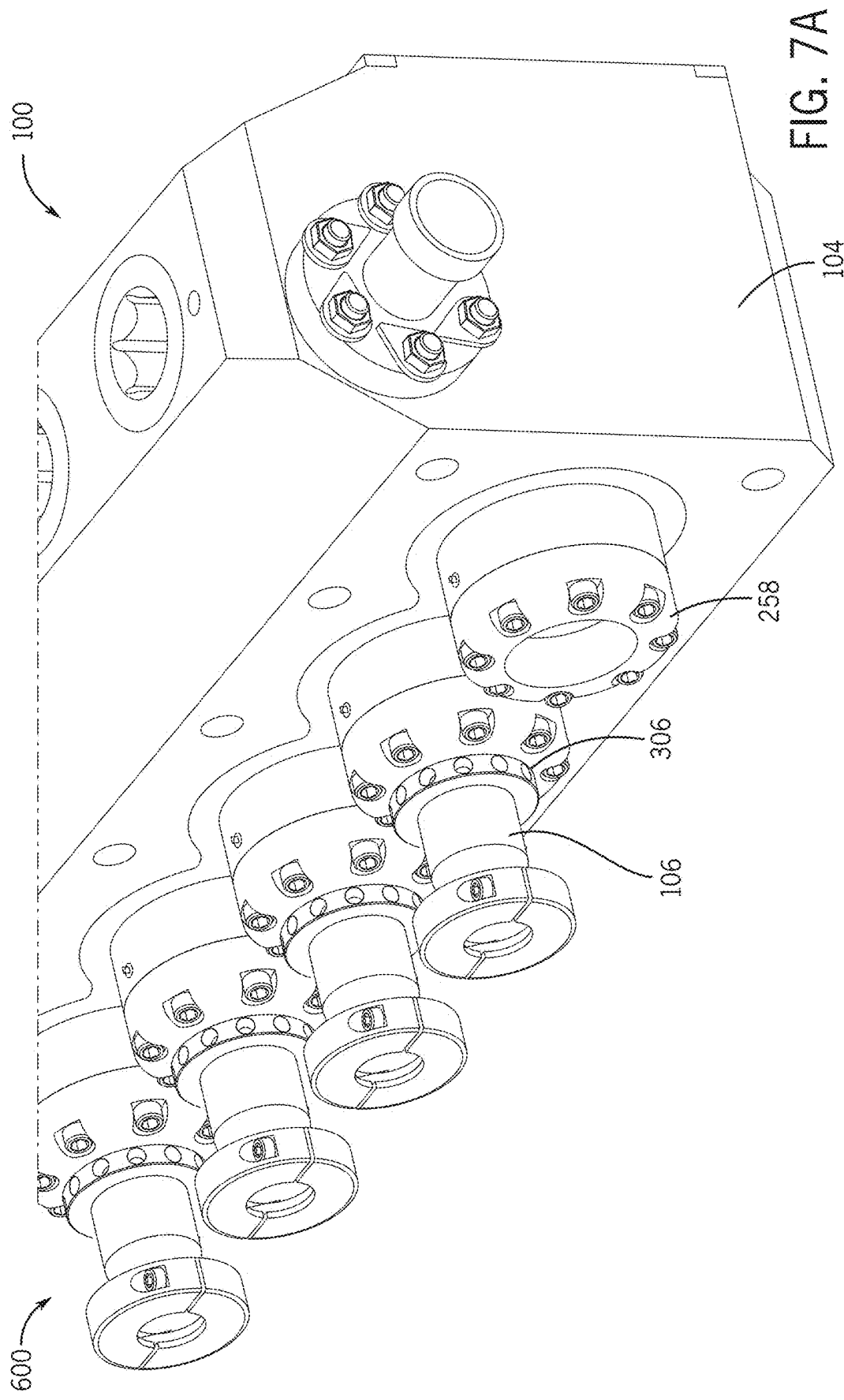
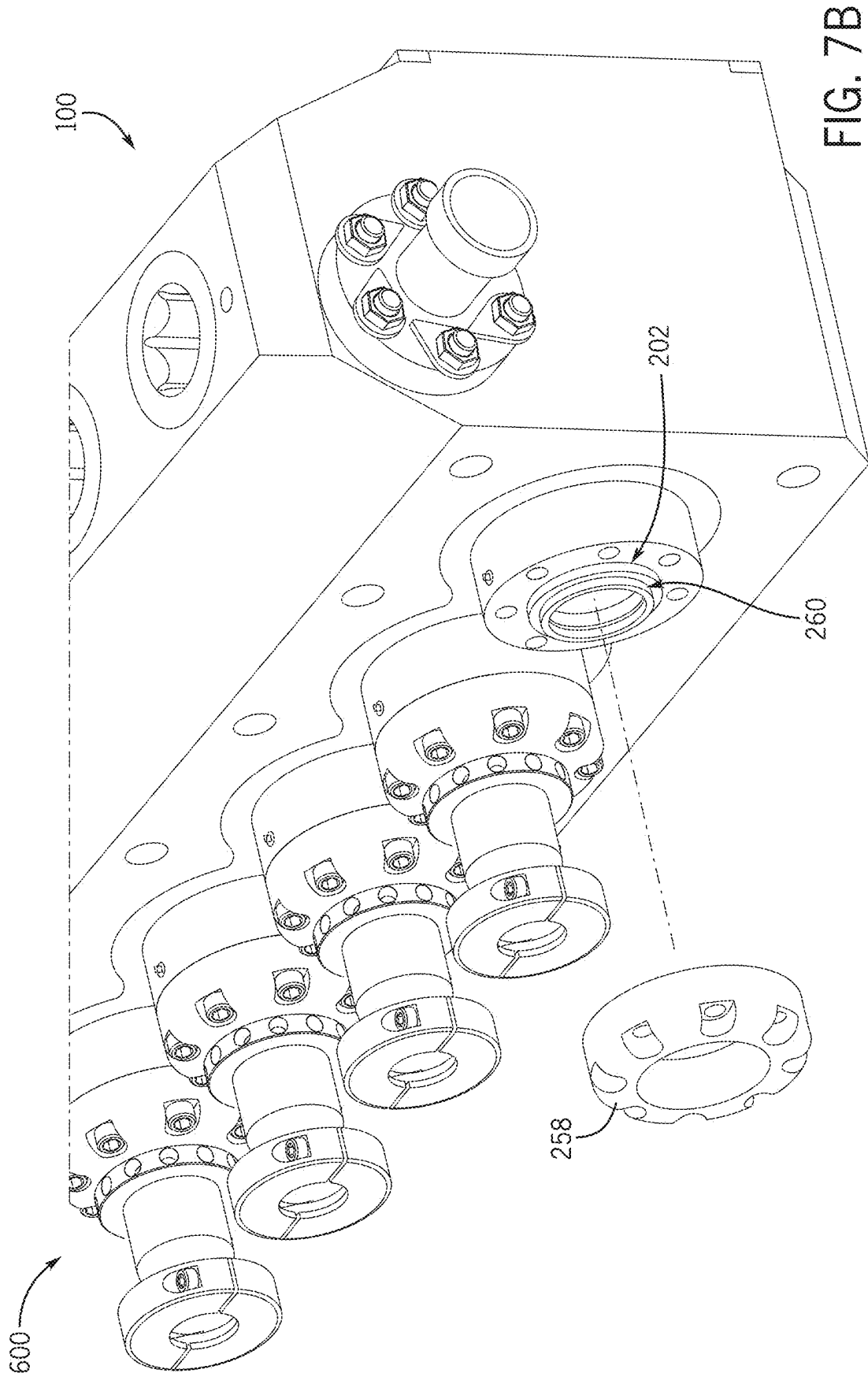


FIG. 6F





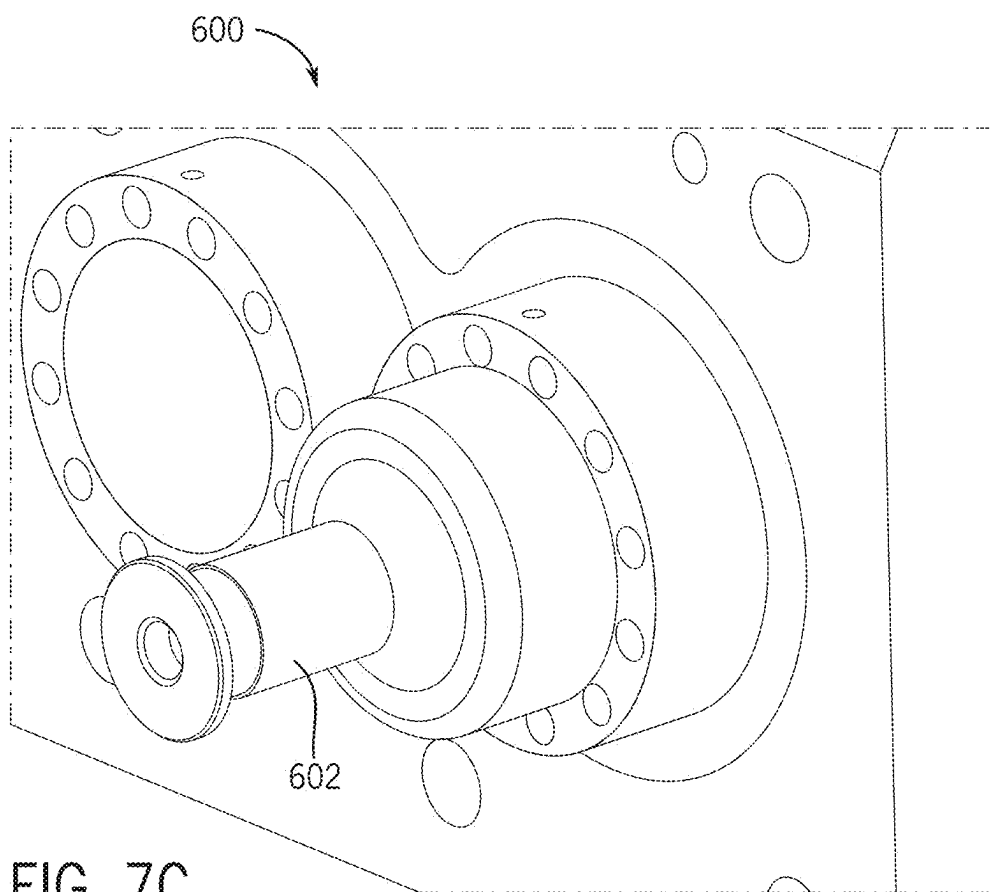
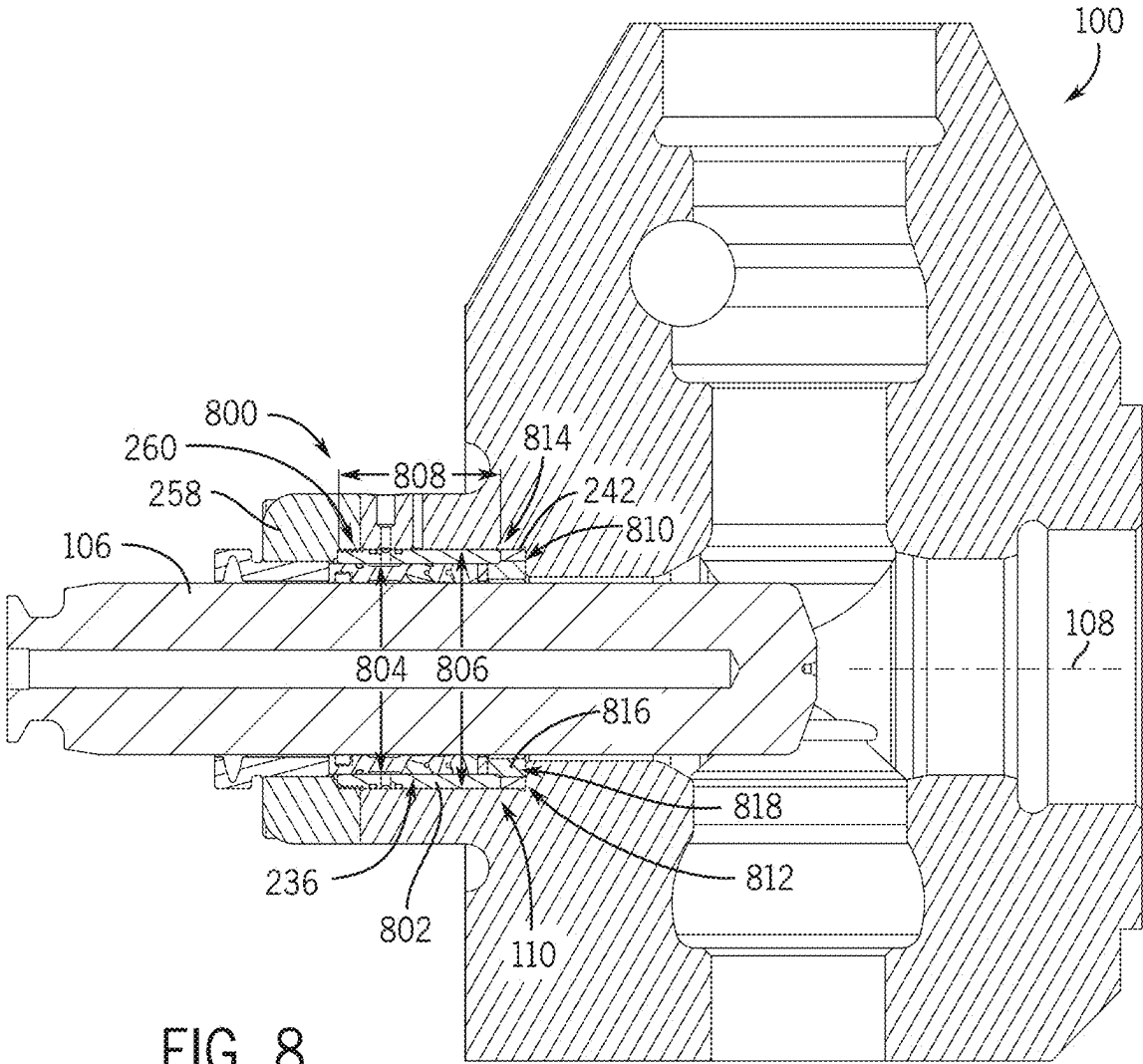
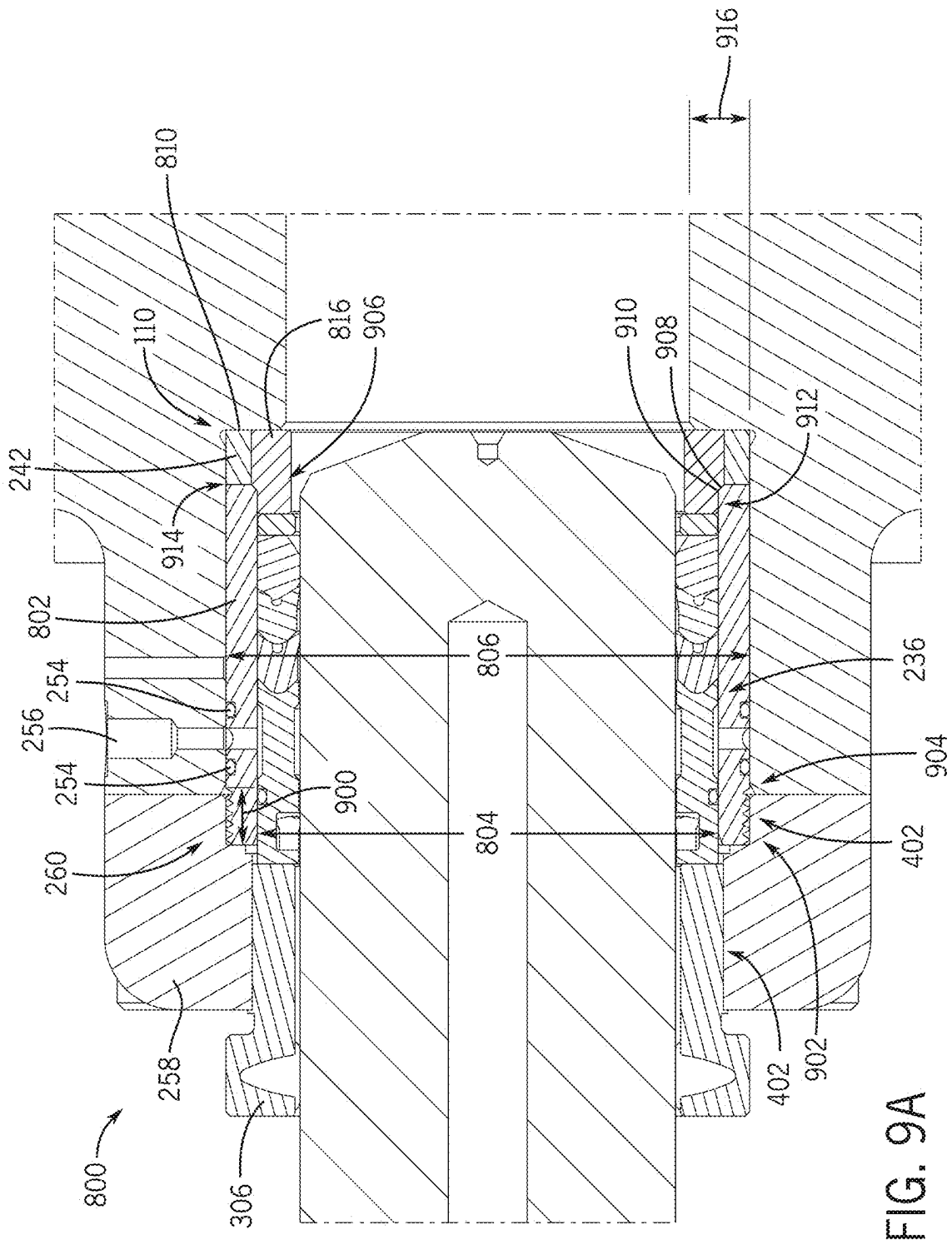
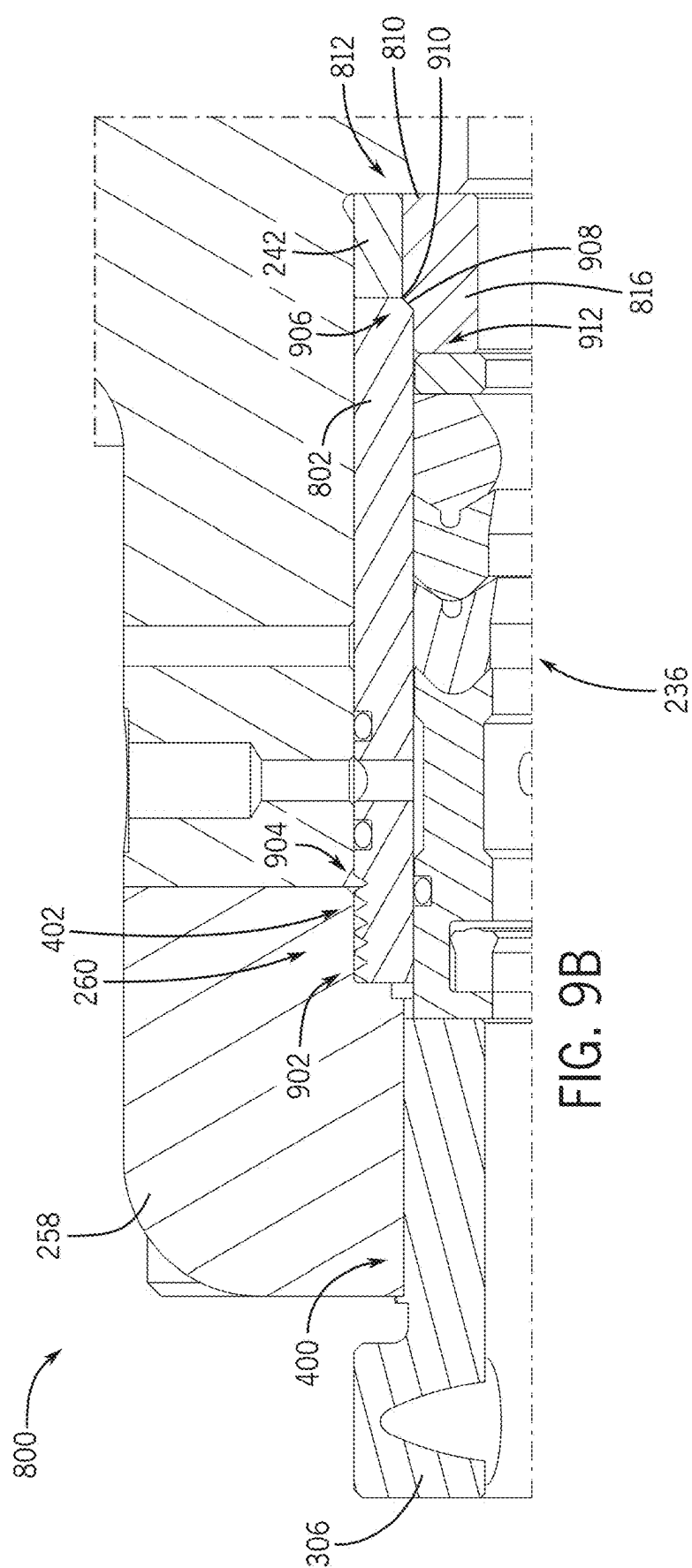


FIG. 7C







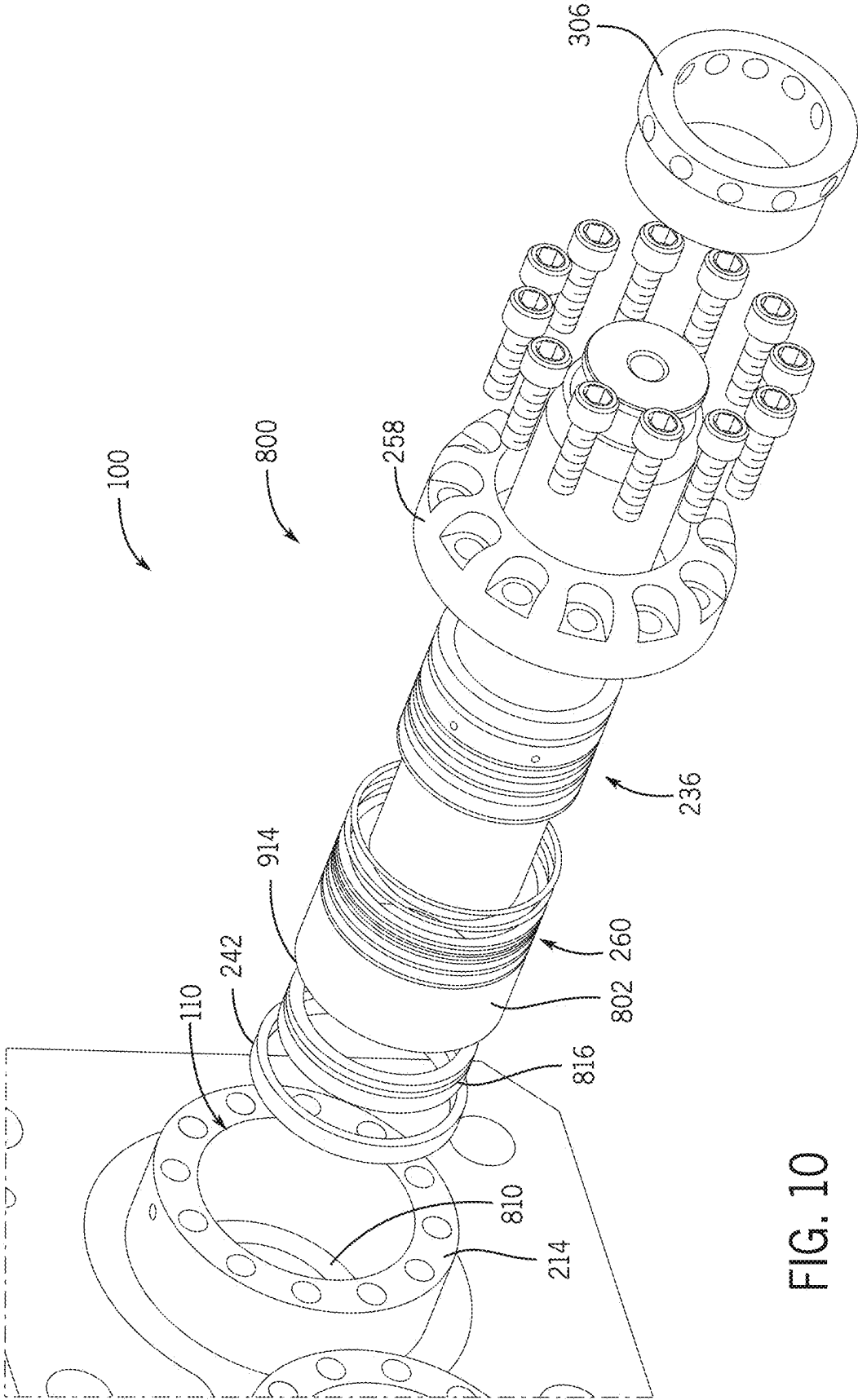
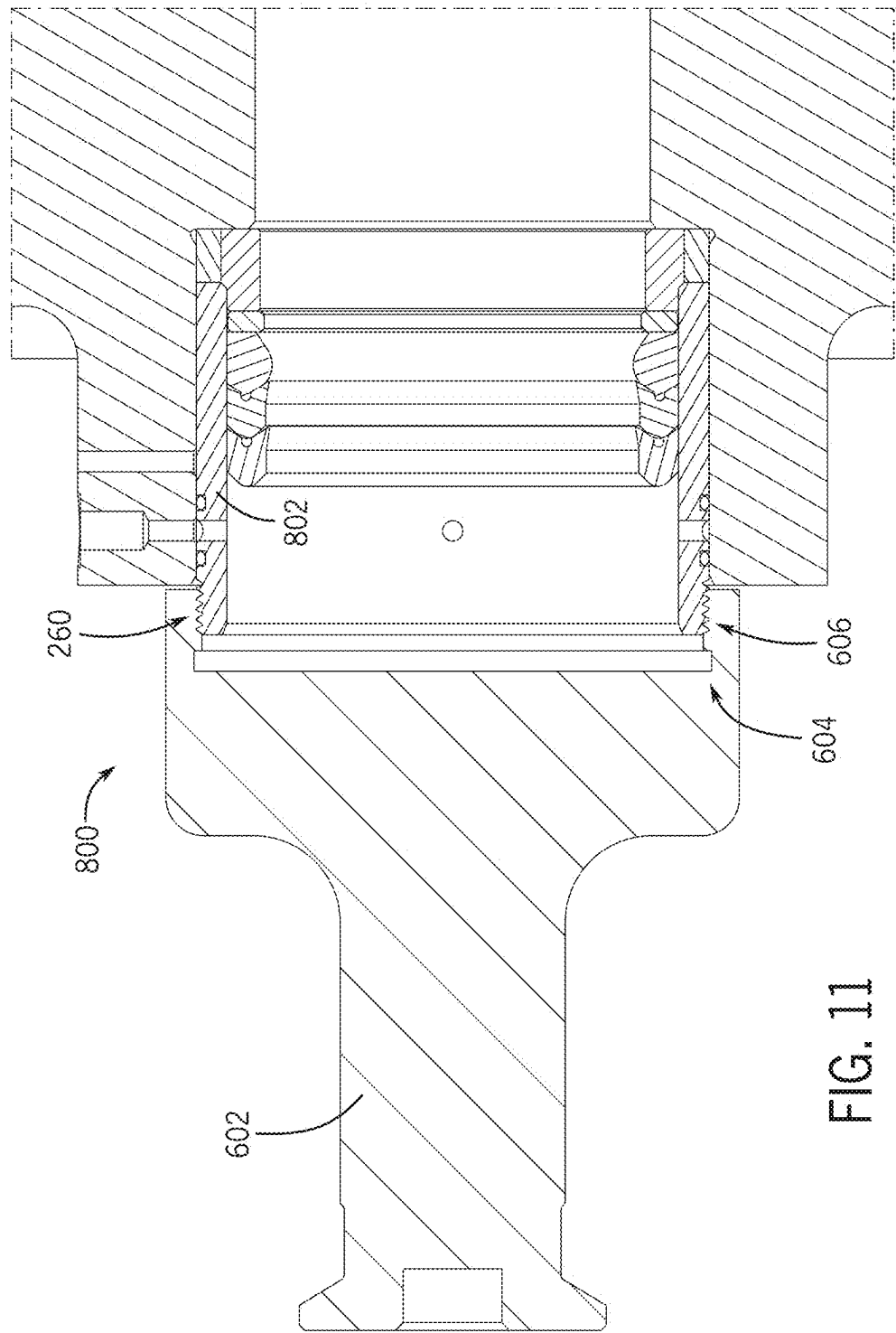


FIG. 10



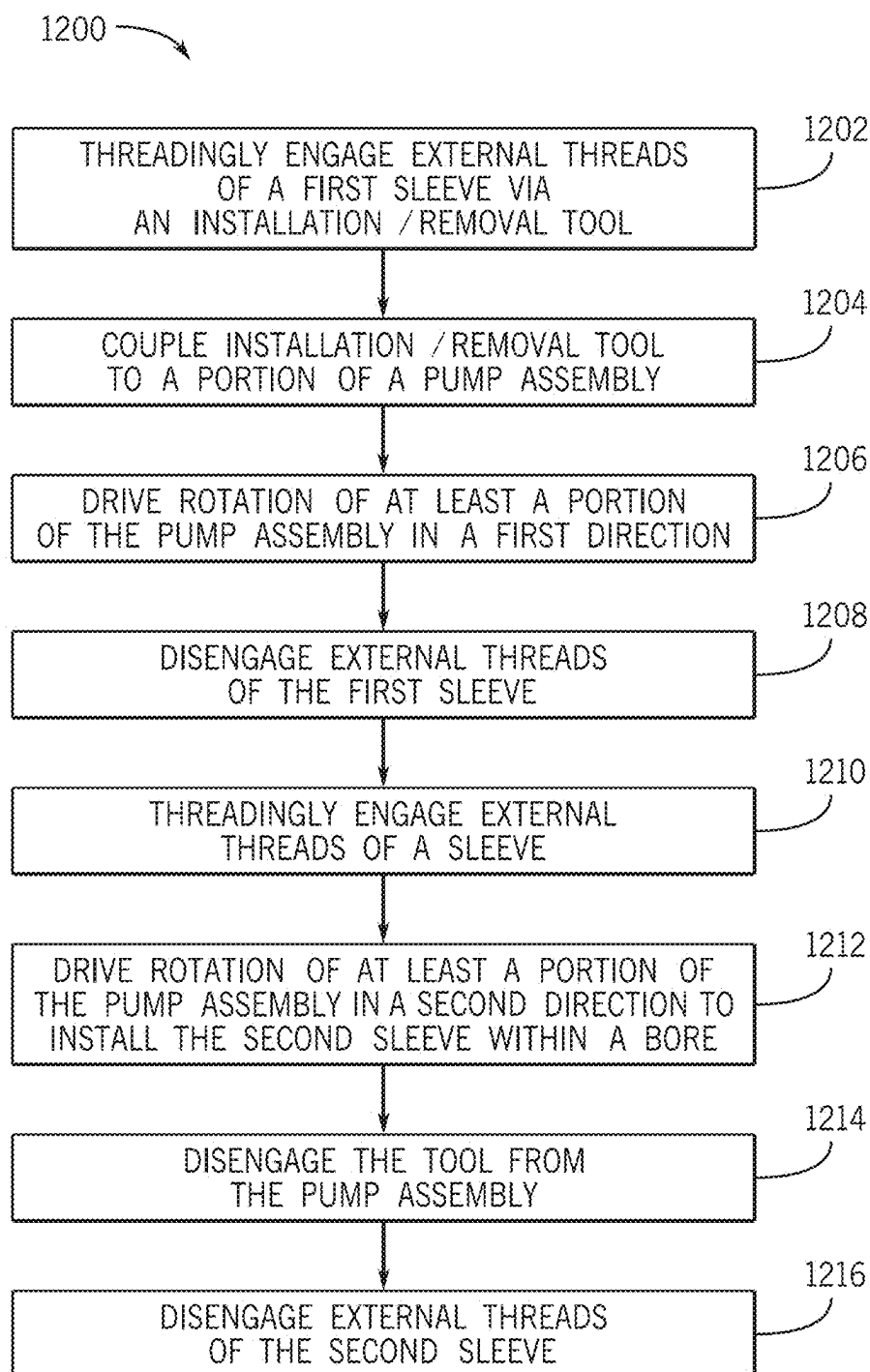


FIG. 12

PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional of U.S. patent application Ser. No. 17/890,818, filed Aug. 18, 2022, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” which is a divisional of U.S. patent application Ser. No. 17/308,163, filed May 5, 2021, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” now U.S. Pat. No. 11,421,679 issued Aug. 23, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 16/916,657, filed Jun. 30, 2020, titled “SLEEVED PACKING BORE SEALING SYSTEM,” the full disclosures of which are hereby incorporated by reference in their entirety for all purposes.

TECHNICAL FIELD

[0002] Embodiments of the subject matter disclosed herein generally relate to pump systems, and in particular to sealing systems.

BACKGROUND

[0003] Pumping systems may be used in a variety of applications, such as industrial applications where pumping systems are used to elevate a working fluid pressure. One such application is hydraulic fracturing systems, where high pressure pumps are used to increase a fluid pressure of a working fluid (e.g., fracturing fluid, slurry, etc.) for injection into an underground formation. The working fluid may include particulates, which are injected into fissures of the formation. When the fluid is removed from the formation, the particulates remain and “prop” open the fissures, facilitating flow of oil and gas. In many applications, reciprocating pumps are used where a fluid is introduced into a fluid end inlet passage and out through an outlet passage. A plunger reciprocates within a bore to add energy to the fluid. Due to the particulates and corrosive nature of the working fluid, various sealing surfaces become eroded or otherwise damaged.

SUMMARY

[0004] Applicants recognized the problems noted above herein and conceived and developed embodiments of systems and methods, according to the present disclosure, for sealing systems.

[0005] In accordance with one or more embodiments, a fluid end includes a first bore extending through a body toward a pressure chamber. The fluid end also includes a second bore extending through the body and intersecting the pressure chamber. The fluid end further includes a packing assembly positioned within the first bore. The packing assembly includes a seal positioned against an inner face of the first bore. The packing assembly also includes a sleeve positioned against the seal, both the sleeve and the seal arranged at a radially outward location of the first bore, the sleeve having radially outwardly facing coupling components at an end opposite the seal. The packing assembly further includes a ring positioned radially inward of the seal.

The packing assembly includes a plurality of packing components positioned on the ring.

[0006] In accordance with another embodiment, a packing assembly includes a packing sleeve having a first end and a second end, the first end having a coupling component and configured to extend, at least partially, axially out of bore of a fluid end block. The packing assembly also includes a seal adapted to sit on a shelf formed within the bore, the seal positioned for engagement by the second end of the packing sleeve. The packing assembly further includes a ring configured to sit on the shelf, radially inward of the seal, at least a portion of the ring axially overlapping the second end along the bore. The packing assembly includes a plurality of packing components configured to sit on a ring shelf, the plurality of packing components to be positioned radially inward of the sleeve and the seal.

[0007] In accordance with another embodiment, a method includes threading a first sleeve arranged within a bore of a pump assembly via a tool. The method also includes coupling the tool to the pump assembly. The method further includes axially removing the first sleeve from the bore, responsive to at least partial first operation of the pump assembly. The method also includes threading a second sleeve to the tool. The method includes axially installing the second sleeve into the bore, responsive to at least partial second operation of the pump assembly.

[0008] In accordance with another embodiment, a fluid end includes a first bore extending through a body toward a pressure chamber. The fluid end also includes a second bore extending through the body and intersecting the pressure chamber. The fluid end further includes a packing assembly positioned within the first bore, the packing assembly having a sleeve that compresses a seal against at least a portion of the first bore to block fluid flow from the first bore, the sleeve extending along a sleeve length, wherein at least a first portion of the sleeve length is positioned within the first bore and at least a second portion of the sleeve length extends axially beyond the first bore, the second portion having a coupling component, and packing components positioned radially inward at least a portion of the sleeve. The fluid end also includes a cap coupled to the body, the cap overlapping the first portion of the sleeve.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

[0010] FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly, in accordance with embodiments of the present disclosure;

[0011] FIG. 2A is a schematic cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0012] FIG. 2B is a detailed cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0013] FIG. 3A is a schematic cross-sectional view of an embodiment of a packing assembly, in accordance with embodiments of the present disclosure;

[0014] FIG. 3B is a detailed cross-sectional view of an embodiment of a packing assembly, in accordance with embodiments of the present disclosure;

[0015] FIG. 3C is a detailed cross-sectional view of an embodiment of a packing assembly, in accordance with embodiments of the present disclosure;

[0016] FIG. 4 is an exploded cross-sectional view of an embodiment of a pump assembly, in accordance with embodiments of the present disclosure;

[0017] FIG. 5 is a cross-sectional view of an embodiment of a sleeve, in accordance with embodiments of the present disclosure;

[0018] FIGS. 6A-6F are cross-sectional views of an embodiment of a sequence for removal of a sleeve, in accordance with embodiments of the present disclosure;

[0019] FIGS. 7A-7C are isometric views of an embodiment of a sequence for removal of a sleeve, in accordance with embodiments of the present disclosure;

[0020] FIG. 8 is a schematic cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0021] FIG. 9A is a detailed cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0022] FIG. 9B is a detailed cross-sectional view of an embodiment of a pump assembly including a packing assembly, in accordance with embodiments of the present disclosure;

[0023] FIG. 10 is an exploded view of an embodiment of a pump assembly, in accordance with embodiments of the present disclosure;

[0024] FIG. 11 is a cross-sectional view of an embodiment of a sleeve coupled to a tool, in accordance with embodiments of the present disclosure; and

[0025] FIG. 12 is a flow chart of an embodiment of a method for sleeve installation and removal, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0026] The foregoing aspects, features, and advantages of the present disclosure will be further appreciated when considered with reference to the following description of embodiments and accompanying drawings. In describing the embodiments of the disclosure illustrated in the appended drawings, specific terminology will be used for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

[0027] When introducing elements of various embodiments of the present disclosure, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, “certain embodiments”, or “other embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodi-

ments that also incorporate the recited features. Furthermore, reference to terms such as “above”, “below”, “upper”, “lower”, “side”, “front”, “back”, or other terms regarding orientation or direction are made with reference to the illustrated embodiments and are not intended to be limiting or exclude other orientations or directions.

[0028] Embodiments of the present disclosure are directed two one or more sealing systems for pumps, which may include reciprocating pumps. In at least one embodiment, one or more sealing systems may further be incorporated in various systems for pressure containment, and it should be appreciated that descriptions with respect to pumping systems or pumps is by way of example only and not intended to limit the scope of the present disclosure. Embodiments may be directed toward a sleeve (e.g., sleeve assembly, packing sleeve, insert, etc.) that includes external threads positioned for engagement by an installation/retrieval tool. In at least one embodiment, external threads are not engaged with other components of the pumping system during normal pumping operations. Accordingly, systems and methods may be directed toward a sleeve that interacts with an installation tool to facilitate rapid installation and removal of one or more components of a sealing system.

[0029] In at least one embodiment, systems and methods may be directed toward an installation tool that threadingly engages one or more sleeves, which may include mating threads as noted above. The installation tool may further be coupled to one or more pump pony rods via a clamp or other reasonable connection. In at least one embodiment, the installation tool may be used for rapid removal and installation of one or more sealing systems. By way of example only, a sleeve may be removed by threading or otherwise coupling a tool to the sleeve and engaging the tool to the pump, as noted above. Driving rotation of the pump crankshaft may facilitate removal of the sleeve. Thereafter, the sleeve may be threaded and a replacement may be coupled to the tool. Subsequent rotation of the crankshaft in an opposite direction may then be used to install the replacement sleeve.

[0030] In at least one embodiment, the sleeve may, at least in part secure a portion of packing or other sealing material within the a pump bore. For example, the seal may be positioned axially next to the packing and installation may compress the sleeve against the packing. In various embodiments, the sleeve may include one or more shelves to receive at least a portion of the packing material. In at least one embodiment, the sleeve may include an extension or arm to compress or otherwise engage packing material. In at least one embodiment, the sleeve may include combinations of elements to enable installation in a variety of different configurations. By way of example only, the sleeve may include a shelf, but the shelf may not be used in configurations where the sleeve is used to axially compress the packing. Moreover, the shelf may be used as the extension and not to receive the packing material.

[0031] FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly 100, which may also be referred to as a reciprocating pump assembly and/or a reciprocating pump. The pump assembly 100 may be utilized during hydraulic fracturing operations, among other operations, where a working fluid (e.g., fracturing fluid, slurry, etc.) is introduced into the pump and energy is added to the working fluid to increase a pressure of the working fluid. Fracturing fluid, by way of example only, may include

corrosives and also particulates, such as sand or ceramics, which are utilized during fracturing operations. These corrosives and particulates cause erosion within the pump assembly 100, which may undesirably affect fracturing operations and lead to down times to replace various components. Additionally, the fracturing fluids may include corrosive acids and the like, which may wear down components of the pump assembly 100.

[0032] It should be appreciated that various components of the pump assembly 100 have been removed for clarity with the following discussion. For example, a power end has been removed in favor of focusing on the illustrated fluid end 102 of the pump assembly 100. The power end may include a crankshaft that is driven by an engine or motor to facilitate operations. The fluid end 102 includes a fluid end block 104 that may house one or more components discussed herein. A plunger rod 106 is driven (e.g., via the crankshaft) to reciprocate within the fluid end block 104 along a plunger axis 108. The plunger rod 106 is positioned within a bore 110 extending through at least a portion of the fluid end block 104. The illustrated bore 110 is arranged along the plunger axis 108 (e.g., first axis) and intersects a pressure chamber 112, which is arranged along a pressure chamber axis 114 (e.g., second axis), which is positioned substantially perpendicular to the plunger axis 108. It should be appreciated that the pump assembly 100 may include multiple plunger rod and pressure chamber arrangements, which may be referred to as a plunger throw. For example, the pump assembly 100 may be a triplex pump, quadplex pump, quintuplex pump, and the like.

[0033] The illustrated fluid end block 104 includes an inlet passage 116 and an outlet chamber 118, which are generally coaxial and arranged along the pressure chamber axis 114. In other words, the inlet passage 116 and the outlet chamber 118 are axially aligned with respect to one another and/or the pressure chamber 112. In various embodiments, fluid enters the pressure chamber 112 via the inlet passage 116, for example on an up stroke of the plunger rod 106, and is driven out of the pressure chamber 112 an outlet passage 120, for example on a down stroke of the plunger 106.

[0034] Respective valve assemblies 122, 124 are arranged within the inlet passage 116 and the outlet chamber 118. These valve assemblies 122, 124 are spring loaded in the illustrated embodiment, but it should be appreciated that such an arrangement is for illustrative purposes only. In operation, a differential pressure may drive movement of the valve assemblies. For example, as the plunger rod 106 is on the upstroke, pressure at the inlet passage 116 may overcome the spring force of the valve assembly 122, thereby driving fluid into the pressure chamber 112. However, on the down stroke, the valve assembly 122 may be driven to a closed position, while the spring force of the valve assembly 124 is overcome, thereby enabling the fluid to exit via the outlet passage 120.

[0035] In one or more embodiments, a sealing element (not pictured) is arranged about the plunger rod 106 to block high pressure fluid from flow outside of the block 104. In at least one embodiment, one or more shoulders or groove may be formed within the block 104 to receive at least a portion of the sealing elements or components associated with the sealing elements. By way of example, one or more sealing elements may be positioned against a shoulder or groove within a bore and then compressed and/or axially maintained at a location by a sleeve that is secured to the block 104, such

as by fasteners, threads, or the like. In at least one embodiment, installation and removal of the sleeve may be a challenging or time consuming process, and accordingly, systems and methods are directed toward sleeves and/or installation/removal tools to facilitate installation and removal of sealing elements.

[0036] FIGS. 2A and 2B are cross-sectional views of the pump assembly 100 where the plunger rod 106 is illustrated within the bore 110. It should be appreciated that the configuration of various components of FIGS. 2A and 2B are by way of example only and are not intended to limit the scope of the present disclosure. As will be described herein, an illustrated sleeve may include a shelf in at least one embodiment, but the shelf may be omitted in other embodiments, as noted herein and further described below. The illustrated bore 110 includes a variable diameter including a stepped portion, which will be described herein, and receives a packing assembly 200, which includes a packing sleeve 202 (e.g., sleeve). The sleeve 202 forms a barrier between a wall 204 of the bore 110 and the plunger rod 106. As will be described below, during operation the sleeve 202 may be subjected to harsh operating conditions and erosion. It may be desirable to form the sleeve 202 as a sacrificial component for replacement over a lifetime of the pump assembly 100. For example, the sleeve 202 may be formed from a different material than the block 104 and/or from a material with altered/enhanced corrosion or erosion resistant properties.

[0037] The illustrated packing assembly 200 is positioned within the bore 110 that includes a first bore section 206, a second bore section 208, and a third bore section 210. It should be appreciated that the three sections 206, 208, 210 are shown for illustrative purposes and that in other embodiments more or fewer sections may be included. The illustrated bore 110 has a variable diameter across an axial length 212 extending from a face 214 toward the pressure chamber 112. In this configuration, a bore diameter decreases from the face 214 to the chamber 112. For example, the first bore section 206 has a first bore diameter 216, the second bore section 208 has a second bore diameter 218, and the third bore section 210 has a third bore diameter 220. The first bore diameter 216 is larger than both the second bore diameter 218 and the third bore diameter 220, while the second bore diameter 218 is also larger than the third bore diameter 220. The variable diameter bore may enable a shelf or shoulder to receive at least a portion of the sleeve 202, which may facilitate locating the sleeve 202 in a desired position and/or provide an indication to an operator for successful installation, among other benefits.

[0038] The sleeve 202 is a substantially cylindrical component having a body 222 with a bore 224 extending through the body 222. The bore 224 is a variable diameter bore 224 and the body 222 includes a first sleeve portion 226 and a second sleeve portion 228, each having a respective diameter 230, 232 (e.g., inner diameter). As noted above, and will be described herein, other embodiments may include a constant or substantially constant diameter bore 224 with both the first sleeve portion 226 and the second sleeve portion 228 having a common diameter 230, or in some embodiments, a common diameter 232. In this embodiment, the first sleeve portion diameter 230 is larger than the second sleeve portion diameter 232. As illustrated, the change in diameters forms a shelf 234 for receiving one or more packing components 236, such as seals, packing brass, and

the like. In at least one embodiment, the shelf **234** is omitted, for example where the diameter is constant. With the inclusion of the packing components **236**, a working diameter of the first sleeve portion is reduced, such that the packing component diameter **238** may be substantially equal to the second sleeve portion diameter **232**. In operation, the plunger **106** reciprocates through the sleeve **202** such that the packing components **236** may contact the plunger **106**.

[0039] As mentioned, in various embodiments the sleeve **202** is arranged within the bore **110** to provide a sacrificial component between the block **104** and the plunger **106**. To prevent leaks and/or significantly reduce a likelihood of leaks, there may be seals and the like arranged between the sleeve **202** and the bore **110**. In this embodiment, the sleeve **202** includes a groove **240** that receives seal **242** in the second sleeve portion **228**. As shown, the groove **240** is formed in a second sleeve portion outer diameter **244**. The seal **242** may be larger than the groove **240** (e.g., extend radially beyond the second sleeve portion outer diameter **244**) such that the seal **242** is compressed against the bore **110** when the sleeve **202** is installed within the bore **110**. In this matter, fluid may be blocked from entering the first bore section **206**. It should be appreciated that, in alternative configurations, the groove **240** may be omitted and the seal **242** may be moved to act as a face seal that abuts an end of the sleeve **202**, as will be described herein.

[0040] Moving toward the face **214**, a transition **244** is illustrated at the variable diameter portion between the first sleeve portion **226** and the second sleeve portion **228**. While the shelf **234** is formed along the inside, a shoulder **246** is illustrated where the outer diameter changes from the second sleeve portion outer diameter **244** to the first sleeve portion outer diameter **250**. This shoulder **246** may engage an extension **252** between the first bore section **206** and the second bore section **208** representative of the change in diameter, which may help seat or otherwise position the sleeve **202** within the bore **110**. In various embodiments, the shoulder **246** blocks axial movement of the sleeve **202** toward the pressure chamber **112**. As noted above, this may enable operators to determine when the sleeve **202** has been positioned at a desired location. The illustrated embodiment includes the shoulder **246** have a substantially right angle, however, it should be appreciated that the shoulder **246** may be tapered or at a less smaller angle (e.g., between 0 degrees and 90 degrees). Furthermore, in one or more embodiments, the extension **252** may receive a face seal or other seal that is compressed by the sleeve **202**, for example in configurations where the transition **244** is omitted.

[0041] It should be appreciated that various additional seals may also be utilized with the illustrated sleeve **202**. For example, in the illustrated embodiment, additional seal arrangements **254** are arranged surrounding a grease portion **256** and also at an interface with a cap **258**. Accordingly, potential leak paths may be blocked by forming respective grooves that receive seals along various different portions of the sleeve **202**.

[0042] In various embodiments, sleeve **202** may further included threads or coupling components **260**, as illustrated in FIGS. 2A and 2B. In this example, the coupling components **260** extend beyond the face **214**, and in this case are positioned within the cap **258**. In other words, the coupling components **260** may be positioned to not engage the block **104** and/or the bore **110**, but rather, to engage components external to the bore **110**. As noted herein, in various embodi-

ments the external components may correspond to one or more installation/removal tools that may engage the sleeve **202** for efficient removal, for example using internal threads of the tool to engage the coupling components **260** of the sleeve **202**. It should be appreciated that threads are provided as an example for the coupling components **260**, and in various other embodiments, additional fasteners may also be utilized such as tongue and groove fittings, bayonet style connectors, pegs and dowels, glues/adhesives, and the like.

[0043] FIGS. 3A-3C are detailed cross-sectional views of the packing assembly **200** illustrating the sleeve **202** arranged within the bore **110** with the seal **242** positioned within the groove **240** to bear against the block **104** at the second bore section **208**. In other words, the groove **240** and the seal **242** are positioned at the reduced diameter portion of the sleeve **202** corresponding to the second sleeve portion **228**. As noted above, the embodiments shown in FIGS. 3A-3C are for illustrative purposes and, in various embodiments, the groove **240**, as well as the transition **246**, may be omitted in favor of a substantially constant diameter sleeve **202** that compresses the seal **242** against the fluid end block **104**.

[0044] As described above, the sleeve **202** extends into the bore **110** such that at least a portion of the second sleeve portion **228** is positioned within at least a portion of the second bore section **208**. Further axial movement of the sleeve **202** toward the third bore section **210** may be blocked via engagement between the shoulder **248** and the extension **252**. That is, incorporating variable diameter portions of both the bore **110** and the sleeve **202** may enable operators to quickly and efficiently install components because alignment may be indicated to the operator via contact between components. It should be appreciated that the location of the extension **252** being internal to the block **104** (e.g., within the bore sections) is for illustrative purpose and that, in some instances, the extension **252** may be on the outside of the block **104**.

[0045] The groove **240** is formed within the second sleeve portion **228** and extends a groove depth **300** toward the axis **108** such that a groove diameter **302** is less than a second sleeve portion outer diameter **244**. While the illustrated groove **240** is substantially square or rectangular (e.g., parallel sidewalls with a substantially perpendicular base), it should be appreciated that this arrangement is for illustrative purposes only and that, in other embodiments, the groove **240** may have arcuate sidewalls, slanted sidewalls, a single slanted sidewall, or any other reasonable combination. Furthermore, in one or more embodiments, the groove **240** may be omitted. In various embodiments, a seal thickness **304** may be greater than the groove depth **300**, such that the seal **242** extends radially outward from the groove **240**. As will be appreciated, when installed, the seal **242** may be compressed via contact with the end block **104**, which may cause the seal **242** to expand or otherwise fill different portions of the groove **240**. In various embodiments, the seal **242** is formed from a polymer material, but it should be appreciated that any reasonable material, such as metals, elastomers, synthetics, or combinations thereof may be utilized for the seal **242**. Furthermore, while a single seal **242** is illustrated, multiple seals may be positioned within the groove **240** or multiple grooves **240**.

[0046] The second sleeve portion outer diameter **244** is less than the first sleeve portion outer diameter **250**, as noted above, thereby forming the transition **246** and the shoulder

248. It should be appreciated that the respective sizes of the diameters **244**, **250** may be particularly selected based on expected operating conditions. For example, the diameters **244**, **250** may be selected for strength, pressure containing purposes, longevity, weight, or the like. As noted, in one or more embodiments the transition **246** and the shoulder **248** may be omitted in favor of a substantially constant outer diameter for the sleeve **202**.

[0047] The packing components **236** are illustrated within the sleeve bore **224** and bear against the plunger **106**. In this embodiment, the packing components **236** include seals, packing brass, and the like. A packing nut **306** secures the packing components **236** within the sleeve bore **224**, for example, via coupling to the block **104**. Accordingly, operation of the pump is enabled as the plunger **106** reciprocates through the bore **110**. It should be appreciated that, in other embodiments, the packing components may not rest on the sleeve **202** or portions thereof and spacer rings, or other sealing elements, may be arranged within the bore **110** to receive the packing components.

[0048] As noted above, additional sealing arrangements **254** are positioned along the sleeve **202**, for example around the grease port **256** and/or proximate the cap **258**. As a result, various levels of leak protection are provided, which may reduce the time between maintenance activities. In at least one embodiment, the additional sealing components may extend toward the cap **258** and stop at the coupling components **260**. For example, the coupling components may be utilized to couple to one or more installation and removal tools utilized during one or more maintenance operations. In at least one embodiment, as noted above, the coupling components **260** are positioned to extend axially out of the bore **110** and portions may be at least partially overlapped by the cap **258**.

[0049] It should be appreciated that various components of the sleeve **202** may be particularly selected based on operating conditions. For example, a first sleeve portion length **308** is illustrated as being larger than a second sleeve portion length **310** and larger than a first bore section length **312**. That is, in this embodiment, the first sleeve portion **226** extends outwardly from the face **214**. Furthermore, in this configuration, the second sleeve portion length **310** substantially corresponds to a second bore section length **314**. Additionally, due to the variable diameter portions of the sleeve **202**, different portions may have different thicknesses, which may be selected based on the dimensions of associated components. For example, a first sleeve portion thickness **316** is less than a second sleeve portion thickness **318**, which is also less than a transition thickness **320** due to the overlap of the first and second sleeve portions **226**, **228**, at the transition **246**. In various embodiments, relative thicknesses may be substantially constant along the length of the sleeve **202**.

[0050] FIG. 4 is an exploded cross-sectional view of the pump assembly **100** including the packing assembly **200**. As shown, the illustrated packing assembly **200** is positioned proximate the face **214** and bore **110**. The seal **242** is positioned out of the groove **240**, formed lower (e.g., closer to the face) than the transition **246**. Furthermore, the packing components **236** and plunger **106** are also installed within the bore, through the sleeve **202**. The packing components **236** are secured in place via the packing nut **306**.

[0051] In operation, the seal **242** may be arranged within the groove **240** and then the sleeve **202** may be installed until

the shoulder **248** contacts the extension **252**. In embodiments, coatings or the like may be used to prepare various surfaces. Thereafter, the packing components **236** may be installed, but it should be appreciated that the packing components **236** may also be installed prior to installation of the sleeve **202**. As noted herein, in various embodiments the groove **240** is omitted and the seal **242** may be positioned within the bore **110**, for example on a shelf or spacer ring, and then compressed by the sleeve **202**.

[0052] Further illustrated is the coupling component **260**, which may include threads to facilitate coupling to an installation tool for assembly and/or removal. In this example, the coupling component **260** is positioned at an end of the sleeve **202** extending axially away from the bore **110**. In operation, the coupling component **260** may align with the cap **258**, which in this example is illustrated with a threaded portion **400** over a portion of the cap **258** and a non-threaded portion **402** over another portion of the cap **258**. In various embodiments, the non-threaded portion **402** may align with the coupling component **260**, such that the coupling component **260** is disengaged and/or not coupled to another component during operation. However, the coupling component **260** may be utilized during installation and removal operations. Moreover, in one or more embodiments, the cap **258** may be used to protect or otherwise guard the coupling component **260**. It should be appreciated that, in one or more embodiments, the coupling component **260** may also be configured to couple to the cap **258**.

[0053] FIG. 5 is a cross-sectional view of the sleeve **202** and the seal **242** positioned proximate the bore **110**. As discussed, the groove **240** is formed within the second sleeve portion **228** and has a groove diameter **302** that is less than the second sleeve portion outer diameter **244**. The illustrated seal **242** is sized to fit within the groove **240** and for installation into the bore **110**. However, as noted herein, other embodiments may omit the groove **240** and position the seal **242** at different locations, such as directly into bore **110** axially closer than sleeve **202**. Further illustrated is the transition **246** and the increased transition thickness **320** compared to the first sleeve portion thickness **316** and the second sleeve portion thickness **318**. As noted above, these dimensions may be particularly selected based on anticipated operating conditions. By way of example, it may be desirable for the second sleeve portion **228** to be thicker than the first sleeve portion **226** because the second sleeve portion **226** is closer to the pressure chamber **112** and may experience more exposure to the fluids.

[0054] FIGS. 6A-6F illustrate an installation and removal sequence **600** in which the coupling component **260** is utilized for removal and/or installation of one or more sleeves **202**. In this sequence **600**, an installation/removal tool **602** (e.g., tool, installation and removal tool, etc.) is utilized to engage sleeves **202**, for example via coupling components **260**, for removal and/or installation. In one or more embodiments, installation/removal tool **602** may be coupled to one or more components of the pump assembly **100** to facilitate removal, such as via rotational of the pump assembly crankshaft in a forward direction for installation and rotation in a backward direction for removal.

[0055] FIG. 6A illustrates the pump assembly **100** including the packing assembly **200** having the sleeve **202** as described above. As noted, other embodiments may omit or change various components, but it should be appreciated that one or more steps of the removal and/or installation

procedures may be similar. In at least one embodiment, it is determined that removal of the packing assembly 200 is desired, for example to replace the packing components 236, to inspect the sleeve 202 for damage, or the like. FIG. 6B further illustrates the pump assembly 100 after the rod 106 is removed from the bore 110. In this example, the packing nut 306 and cap 258 remain coupled to the block 104, and moreover, the packing components 236 are still positioned within the bore 110. FIG. 6C illustrates removal of the packing nut 306, which permits removal of the packing components 236, among other elements. In this example, the packing nut 306 may be threaded into the cap 258 such that removal utilizes unthreading the packing nut 306 from the cap 258 while the cap 258 remains engaged with the block 104, for example via one or more fasteners coupling the cap 258 to the block 104.

[0056] Further illustrated in FIG. 6D is the installation/removal tool 602 positioned proximate the block 104. In this example, the tool 602 is axially aligned with the axis 108 such that an engaging end 604 is aligned with the sleeve 202. In one or more embodiments, the engaging end 604 includes mating coupling components 606, which in this example are threads to engage the coupling components 260 of the sleeve 202. As shown, an engaging end diameter 608 is greater than the first sleeve portion diameter 230 to facilitate engagement with the coupling components 260, instead of the packing components 236. In at least one embodiment, the tool 602 further includes a working end 610, opposite the engaging end 604, which includes one or more fittings 612 to facilitate coupling to external drivers, such as the pump assembly 100, hand tools, etc. to drive rotation of the tool 602.

[0057] FIG. 6E illustrates engagement between the tool 602 and the sleeve 202 via the components 260, 606. For example, in embodiments where the components 260, 606 are threads, engagement may be facilitated by rotating the tool 602 about the axis 108, which in certain embodiments may be performed via coupling of the tool 602 to the pump assembly 100 and rotating the crankshaft. In various embodiments, one or more hand tools may be used to engage the fitting 612 to drive rotation. FIG. 6F illustrates axial removal of the packing assembly 200 from the bore 110. In this example, the entire assembly 200 may be removed, which may be an improvement over removing individual components from the bore 110 to facilitate faster installation and removal. Accordingly, a new assembly may be installed, which may have a similar configuration as the removed assembly or may have a different configuration, such as a constant diameter sleeve as described herein.

[0058] FIGS. 7A-7C are perspective views of embodiments of the pump assembly 100 illustrating different portions of the removal sequence 600, including removal of the plunger 106 and packing nut 306 (FIG. 7A), removal of the cap 258 (FIG. 7B), and installation of the tool 602 (FIG. 7C). In these examples, various interactions between components may be seen. By way of example, FIG. 7A illustrates threads for installing the packing nut 306 within the cap 258. Furthermore, fasteners secure the cap 258 to the block 104. FIG. 7B further illustrates removal of the cap 258 to expose the coupling components 260 of the sleeve 202, which in various embodiments do not engage the cap 258 and are utilized for installation and removal, for example via the tool (not pictured in FIG. 7B). However, FIG. 7C illustrates coupling of the tool 602 to the coupling components 260, for example via mating threads, to facilitate removal of the

assembly (not visible). In various embodiments, it should be appreciated that the tool 602 may further be coupled to one or more components to facilitate rotation, but this has been omitted for clarity with the figures.

[0059] FIG. 8 is a cross-sectional view of an embodiment of a pump assembly 100 including a packing assembly 800, which may share one or more features with the packing assembly 200, and it should be appreciated that certain components may share reference numerals for convenience, and such numbering should not be construed as a limitation in configurations utilized with the present disclosure. In this example, the packing assembly 800 includes a sleeve 802, which may share one or more features with the sleeve 202, as noted above. In one or more embodiments, the sleeve 802 extends, at least partially, into the bore 110. The sleeve 802 is substantially aligned along the axis 108 and arranged coaxially within the bore 110. The sleeve 802 is positioned radially outward from the rod 106 and packing components 236 are positioned to bear against both the sleeve 802 and the rod 106.

[0060] In this example, the sleeve 802 has a substantially constant inner diameter 804 and a substantially constant outer diameter 806. It should be appreciated that substantially constant refers to a majority percentage of a sleeve length 808 having an approximately equal diameter. In this example, the sleeve 802 is utilized to compress the seal 242 against an inner face 810. In various embodiments, the inner face 810 is formed, at least in part, on a shelf 812 formed by a transition between diameters within the bore 110, as also described above. The illustrated seal 242 is positioned at a radially outward portion 814 of the shelf 812 and is substantially aligned with the sleeve 802. In operation, the sleeve 802 may be utilized to compress the seal 242 to block fluid leaks around the rod 106.

[0061] In one or more embodiments, a ring 816 is also positioned along the shelf 812, and in this example is positioned at a radially inward portion 818. That is, the ring 816 is arranged between the seal 242 and the rod 106. It should be appreciated that this configuration is for illustrative purposes only and that, in one or more embodiments, at least a portion of the seal may be aligned with the ring 816, may be radially inward of the ring 816, or may be in any other reasonable location. In various embodiments, at least a portion of the ring 816 is utilized to support the packing components 236, as will be described below.

[0062] Further illustrated in FIG. 8 are the coupling components 260, which as shown are threads. In this example, the coupling components 260 extend axially out of the bore 110 and are overlapped, at least partially, by the cap 258. In one or more embodiments, the coupling components 260 are not engaged during normal operation of the pump assembly 100 (e.g., not engaged while the pump assembly is running) and are utilized for installation and removal of the packing assembly 900.

[0063] FIGS. 9A and 9B are detailed cross-sectional views of embodiments of the packing assembly 800. In one or more embodiments, the sleeve 802 is arranged within the bore 110 such to compress the seal 242 against the inner face 810 to block fluid flow from the bore 110. In at least one embodiment, the sleeve 802 includes substantially constant inner and outer diameters 804, 806. In one or more embodiments, the sleeve 802 does not include features to support the packing components 236, which may be supported by the ring 816 positioned radially inward of the seal 242.

[0064] FIGS. 9A and 9B further illustrate the grease port 256 and seal arrangement 254 described below. As shown, both are axially closer to the pressure chamber 112 (FIG. 1) than the coupling components 260, which are shown as threads in the illustrated embodiment. In this example, a coupling component length 900 is shown where a first portion 902 may be considered outside of the bore 110 and a second portion 904 may be considered inside of the bore 110. It should be appreciated that the portion or percentage of the length 900 which is inside or outside may be particularly selected based on expected operating conditions. Moreover, the component length 900 may be adjusted to accommodate various tools. In this example, it is shown that the coupling component 260 is aligned with the non-threaded portion 402 of the cap 258, whereas the threaded portion 400 is utilized to secure the packing nut 306.

[0065] In this example, the seal 242 is arranged axially closer to the pressure chamber 112 (FIG. 1) than the sleeve 802. Moreover, the ring 816 is also positioned axially closer than at least a portion of the sleeve 802. However, an overlapping portion 906 is illustrated between the ring 816 and the sleeve 802. At the overlapping portion 906, a sleeve edge 908 is shown to engage a ring edge 910, where both respective edges 908, 910 are arranged at angles. As a result, at least a portion of the sleeve edge 908 may have a different diameter than the remainder of the sleeve 802. A similar configuration is also illustrated at an opposite end of the sleeve 802 and may be utilized to simplify installation and removal, as well as provided a machined surface, among other benefits. It should be appreciated that this configuration is an example and additional angles or mating edges may be utilized, moreover, a curved or arcuate edge may be used to receive one or more of the sleeve 802 and/or the ring 816. In at least one embodiment, a ring shelf 912 receives and supports the packing components 236, where the ring shelf 912 is axially further away from the pressure chamber 112 (FIG. 1) than a sleeve end 914. In other words, the overlapping portion 906 is axially closer to the pressure chamber 112 (FIG. 1) than the packing components 236 in the illustrated embodiment. It should be appreciated that various other configurations may further be utilized within the scope of the present disclosure.

[0066] Various embodiments illustrate both the seal 242 and the ring 816 positioned on the shelf 812, which is formed at a location within the bore 110 having a changing diameter. It should be appreciated that the seal 242 may be considered a face seal and the shelf 812 may include a machined or sealing surface particularly selected to accommodate a sealing component. A shelf width 916 receives both the seal 242 and the ring 816 and, in certain embodiments, at least a portion of the ring 816 may overlap the shelf 812.

[0067] In operation, the packing assembly 800 is positioned within the bore 110 such that the sleeve 802 compresses the seal 242. In one or more embodiments, the seal 242 may be installed within the bore 110 prior to the sleeve 802. By way of example only, the seal 242 may be installed, followed by the ring 816, followed by the sleeve 802, and then the packing components 236. It should be appreciated that this installation procedure is by way of example only and not intended to limit the scope of the present disclosure.

[0068] FIG. 10 is an exploded cross-sectional view of the pump assembly 100 including the packing assembly 800. As shown, the illustrated packing assembly 800 is positioned

proximate the face 214 and bore 110. The seal 242 is positioned out of the bore 110 and axially closer to the inner face 810 than the sleeve 802. Furthermore, the packing components 236 and plunger 106 are also installed within the bore 110, through the sleeve 802. The packing components 236 are secured in place via the packing nut 306.

[0069] In operation, the seal 242 may be arranged within bore 110 against the inner face 810 and then the sleeve 802 may be installed until the sleeve end 914 contacts the seal 242. In embodiments, coatings or the like may be used to prepare various surfaces. Additionally, in embodiments, the ring 816 is also positioned against the inner face 810. Thereafter, the packing components 236 may be installed, but it should be appreciated that the packing components 236 may also be installed prior to installation of the sleeve 802.

[0070] Further illustrated is the coupling component 260, which may include threads to facilitate coupling to an installation tool for assembly and/or removal. In this example, the coupling component 260 is positioned at an end of the sleeve 802 extending axially away from the bore 110. In operation, the coupling component 260 may align with the cap 258, for example with a non-threaded portion (not pictured). In one or more embodiments, the cap 258 may be used to protect or otherwise guard the coupling component 260. It should be appreciated that, in one or more embodiments, the coupling component 260 may also be configured to couple to the cap 258. Moreover, while not illustrated in FIG. 10, installation and/or removal may be accomplished using one or more tools, as described herein.

[0071] FIG. 11 is a cross-sectional view of an embodiment of the packing assembly 800 coupled to the installation/removal tool 602. In this example, the mating coupling components 606 of the engaging end 604 are coupled to the coupling components 260 of the sleeve 802 to facilitate installation and removal. As noted above, in various embodiments, the tool 602 may be coupled to one or more components of the assembly to facilitate removal, for example by driving rotation of the tool 602 and/or by driving axial movement of the tool 602.

[0072] FIG. 12 is a flow chart of an embodiment of a method 1200 for installation and removal of a sleeve. It should be appreciated for this method, and all methods described herein, that there may be more or fewer steps. Additionally, the steps may be performed in a different order, or in parallel, unless otherwise specifically stated. In this example, external threads of a first sleeve are engaged via an installation and removal tool 1202. For example, external threads may correspond to a coupling component on a sleeve where the coupling component is not engaged during operation of the pump but engaged during installation and/or removal. Furthermore, it should be appreciated that while threads are discussed with respect to the method, various other coupling mechanisms may be used in other embodiments of the present disclosure. The tool may be coupled to a portion of a pump assembly 1204. In at least one embodiment, the tool is coupled to pump pony rods via a clamp connection, but other locations may also be used. Rotation of a portion of the pump, such as the crankshaft, may be driven in a first direction 1206. In at least one embodiment, rotation axially moves the tool, and thereafter the coupled sleeve, axially away and out of a bore. The first sleeve is disengaged from the tool 1208 and external threads of a second sleeve are engaged 1210. Rotation of the portion

of the pump may then be applied in a second, opposite, direction to axially move the second sleeve toward the bore **1212**. The tool may then be disengaged from the pump assembly **1214** and the tool may also be disengaged from the second sleeve **1216**. In this manner, one or more sleeves may be removed or installed within bores of a pump assembly.

[0073] This application is a divisional of U.S. patent application Ser. No. 17/890,818, filed Aug. 18, 2022, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” which is a divisional of U.S. patent application Ser. No. 17/308,163, filed May 5, 2021, titled “PACKING ASSEMBLY WITH THREADED SLEEVE FOR INTERACTION WITH AN INSTALLATION TOOL,” now U.S. Pat. No. 11,421,679 issued Aug. 23, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 16/916,657, filed Jun. 30, 2020, titled “SLEEVED PACKING BORE SEALING SYSTEM,” the full disclosures of which are hereby incorporated by reference in their entirety for all purposes.

[0074] The foregoing disclosure and description of the disclosed embodiments is illustrative and explanatory of the embodiments of the disclosure. Various changes in the details of the illustrated embodiments can be made within the scope of the appended claims without departing from the true spirit of the disclosure. The embodiments of the present disclosure should only be limited by the following claims and their legal equivalents.

1. A packing assembly to be positioned within a bore, comprising:

a sleeve to compress a seal against at least a portion of the bore to block fluid flow from the bore, the sleeve positioned to extend along a sleeve length, at least a first portion of the sleeve length positioned within the bore and at least a second portion of the sleeve length positioned to extend axially beyond the bore, the second portion of the sleeve including a coupling component; and

packing components positioned radially inward of at least a portion of the sleeve.

2. The packing assembly of claim 1, wherein the coupling component is positioned on a radially outward facing portion of the sleeve.

3. The packing assembly of claim 1, further comprising a ring to be positioned within the bore, the ring to at least partially compress the seal.

4. The packing assembly of claim 3, wherein the packing components are positioned on a ring shelf.

5. The packing assembly of claim 1, wherein the coupling component is not engaged with the bore.

6. A packing assembly to be positioned within a bore, comprising:

a sleeve to compress a seal against at least a portion of the bore to block fluid flow from the bore, the sleeve extending along a sleeve length, wherein at least a first portion of the sleeve length is positioned within the bore and at least a second portion of the sleeve length extends axially beyond the bore, the second portion of the sleeve having a coupling component positioned on a radially outward facing portion of the sleeve;

a ring positioned within the bore to at least partially compress the seal; and

packing components positioned radially inward of at least a portion of the sleeve.

7. The packing assembly of claim 1, wherein the packing components are positioned on a ring shelf.

8. The packing assembly of claim 1, wherein the coupling component is not engaged with the bore.

9. A packing assembly to be positioned within a bore, comprising:

a sleeve to compress a seal against at least a portion of the bore to block fluid flow from the bore, the sleeve positioned to extend along a sleeve length, at least a first portion of the sleeve length positioned within the bore and at least a second portion of the sleeve length positioned to extend axially beyond the bore, the second portion of the sleeve including a coupling component;

a ring positioned within the bore; and

packing components positioned radially inward of at least a portion of the sleeve and located on a ring shelf.

10. The packing assembly of claim 1, wherein the coupling component is positioned on a radially outward facing portion of the sleeve.

11. The packing assembly of claim 1, wherein the ring is positioned to at least partially compress the seal.

12. The packing assembly of claim 1, wherein the coupling component is not engaged with the bore.

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